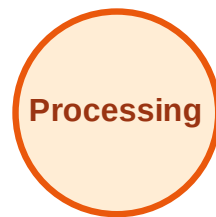


# BFS Traversal

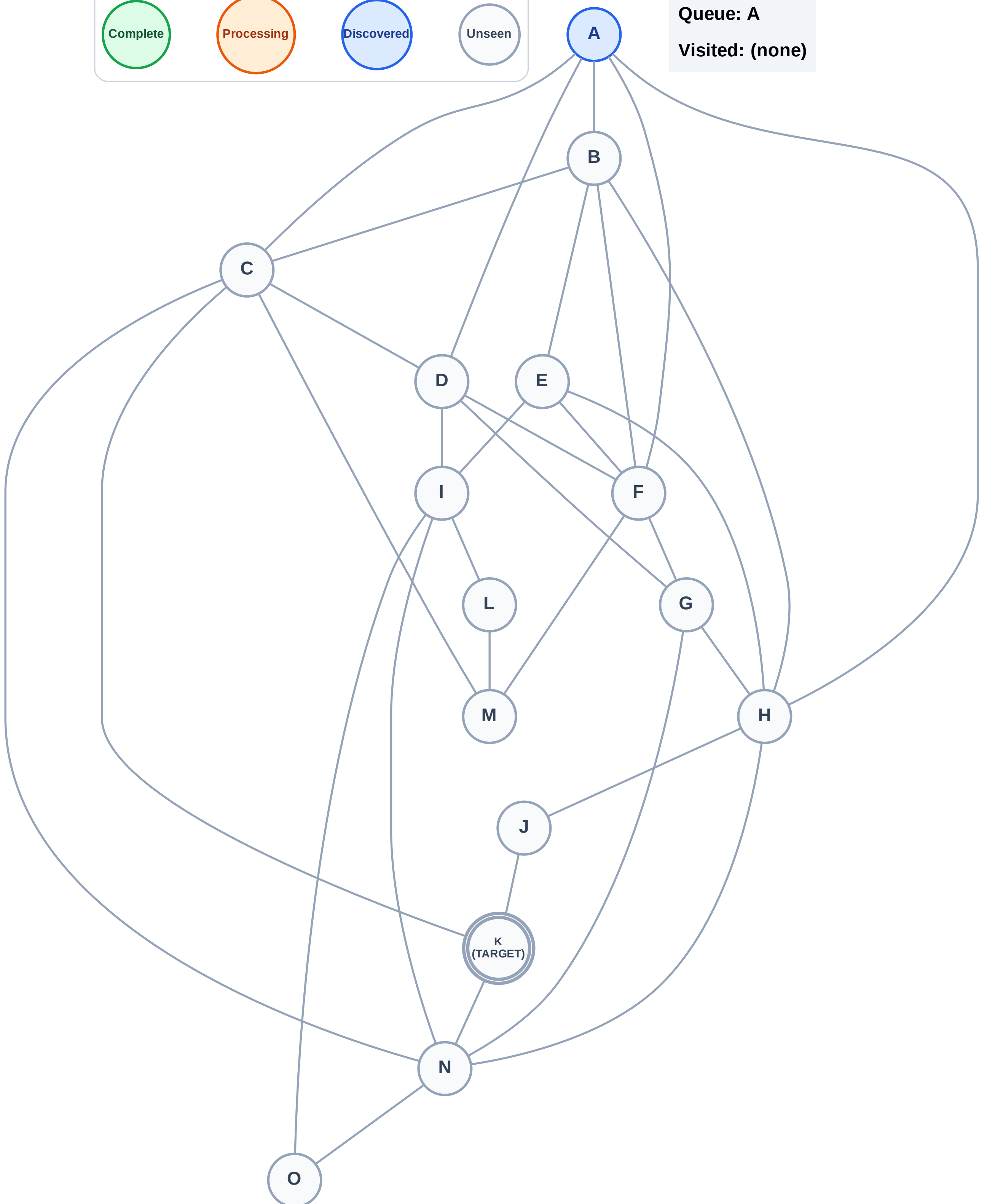
Step 1: Start at A (Seeking target node K)

## Legend



Queue: A

Visited: (none)



# BFS Traversal

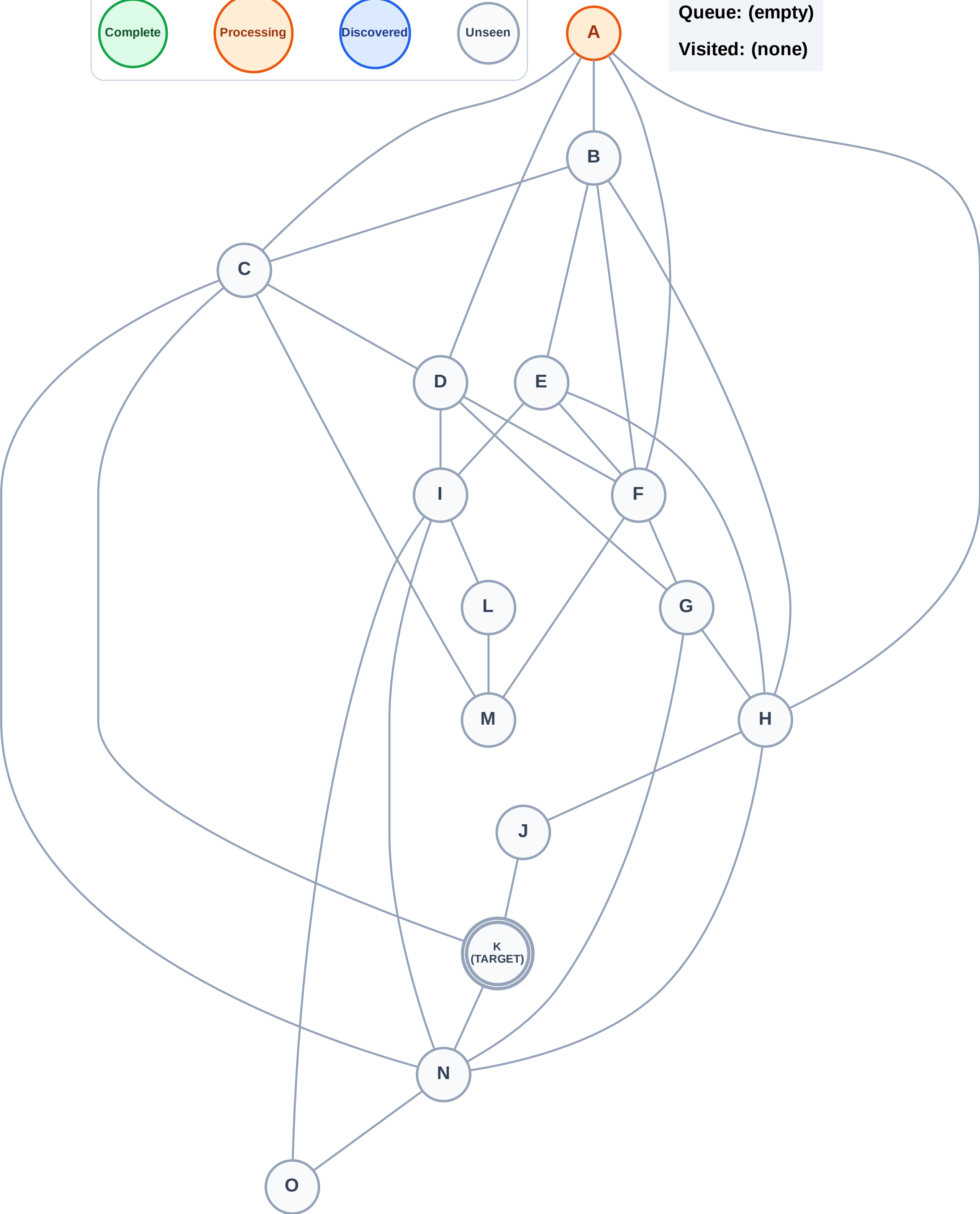
Step 2: Process node A

Legend



Queue: (empty)

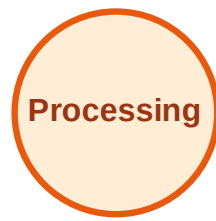
Visited: (none)



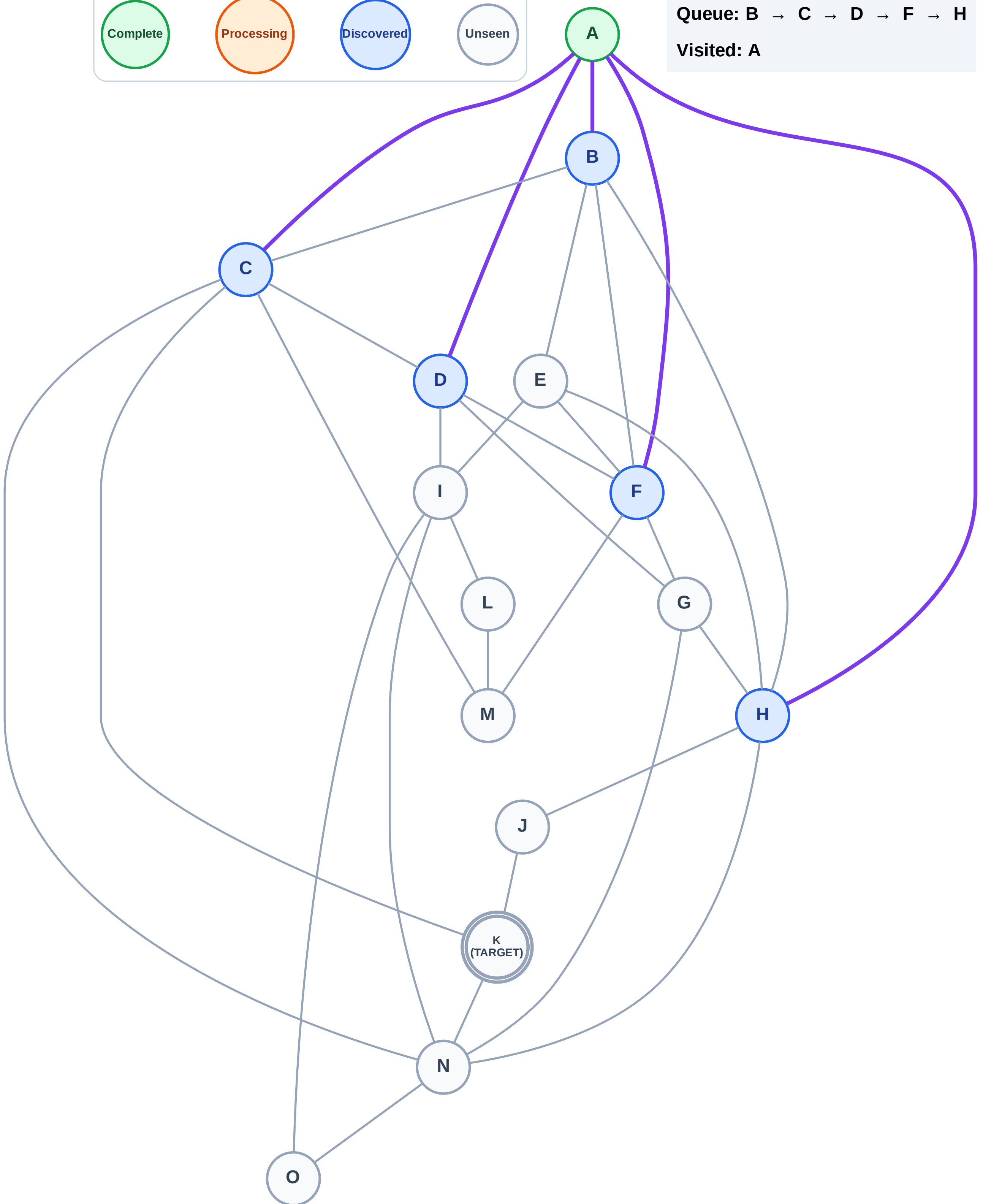
# BFS Traversal

Step 3: Discover B, C, D, F, H from A

## Legend



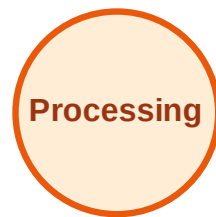
Queue: B → C → D → F → H  
Visited: A



# BFS Traversal

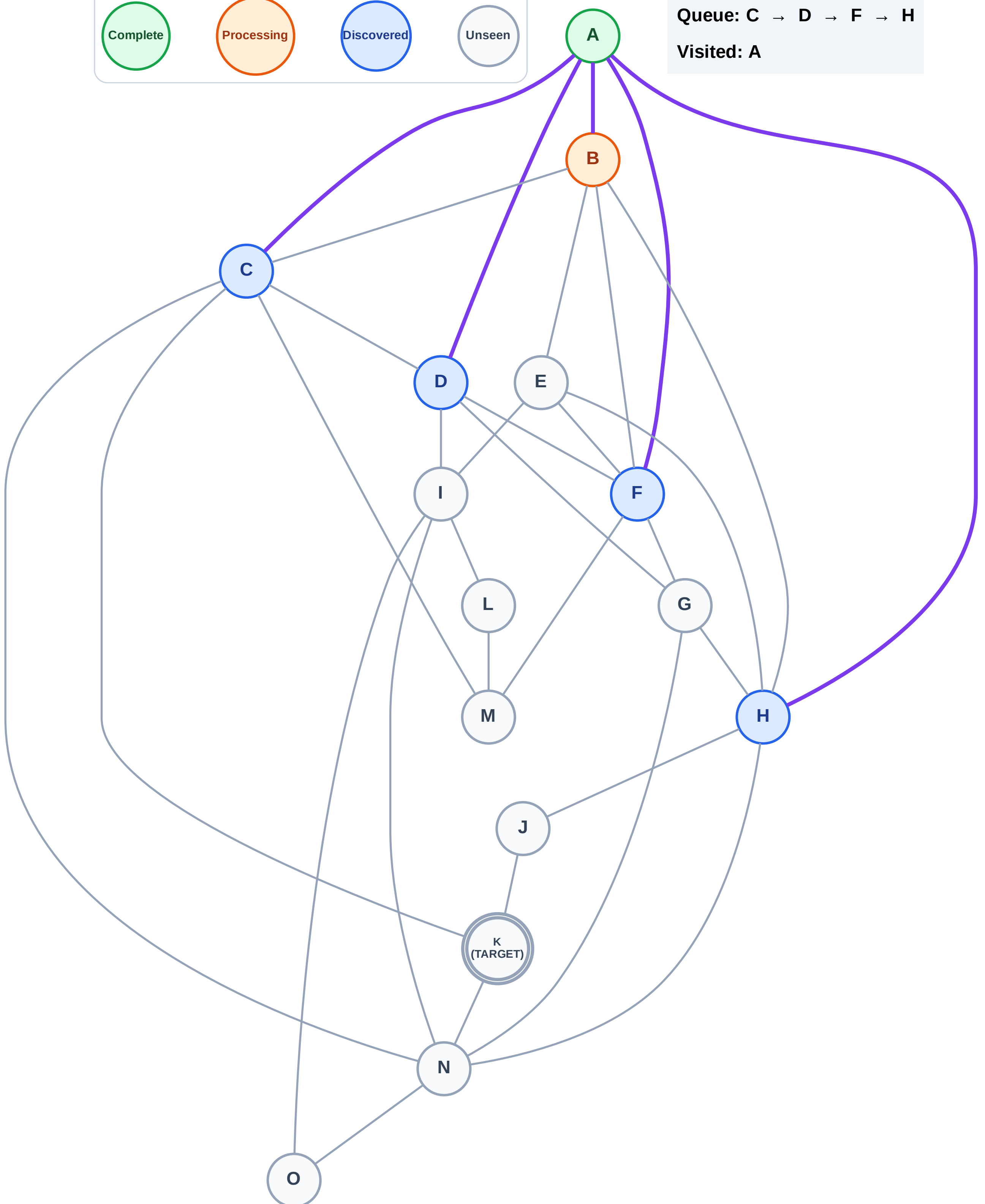
Step 4: Process node B

## Legend



Queue: C → D → F → H

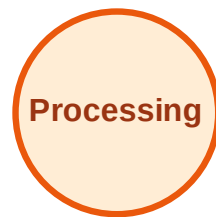
Visited: A



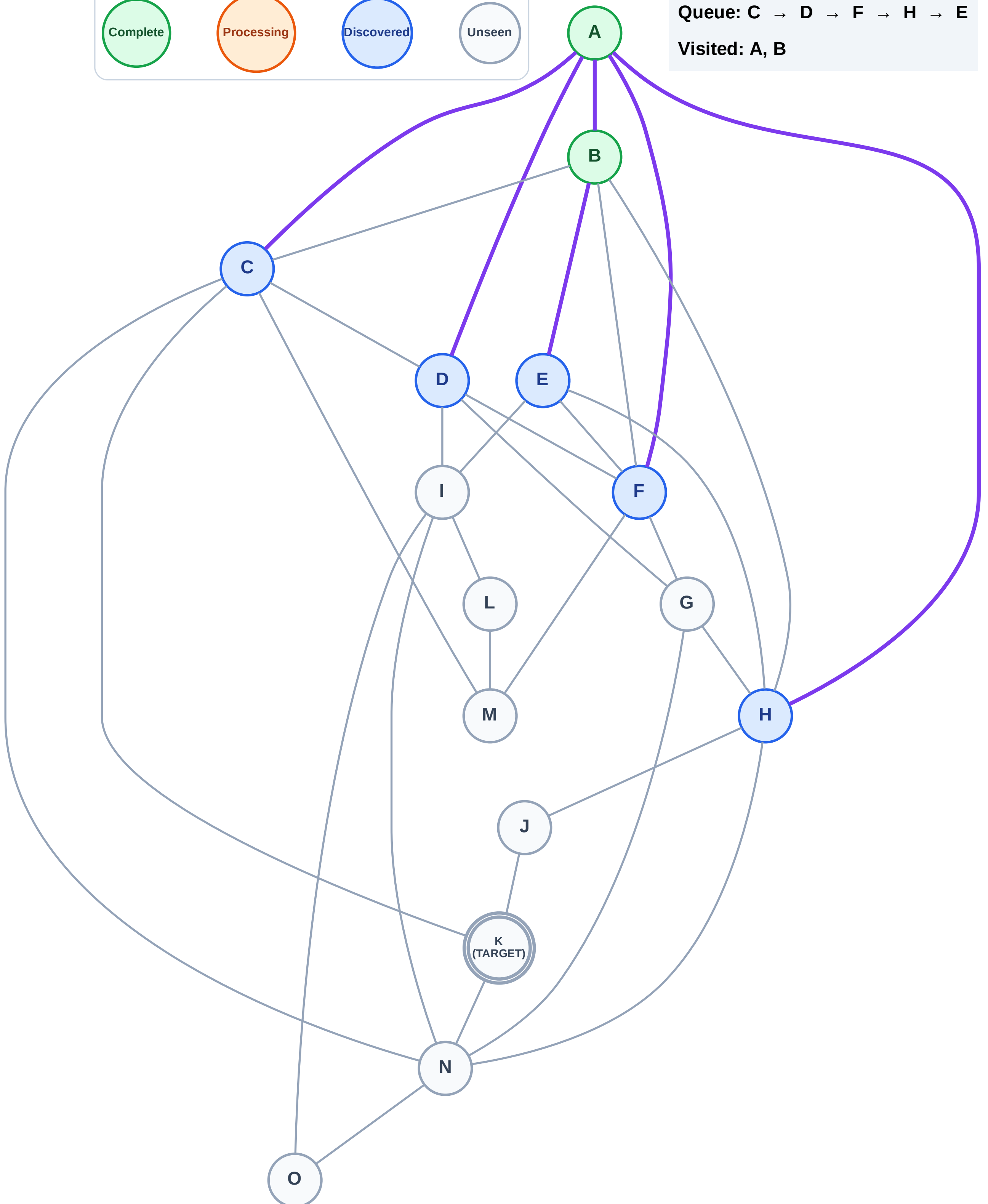
# BFS Traversal

Step 5: Discover E from B

## Legend



Queue: C → D → F → H → E  
Visited: A, B



# BFS Traversal

Step 6: Process node C

Legend

Complete

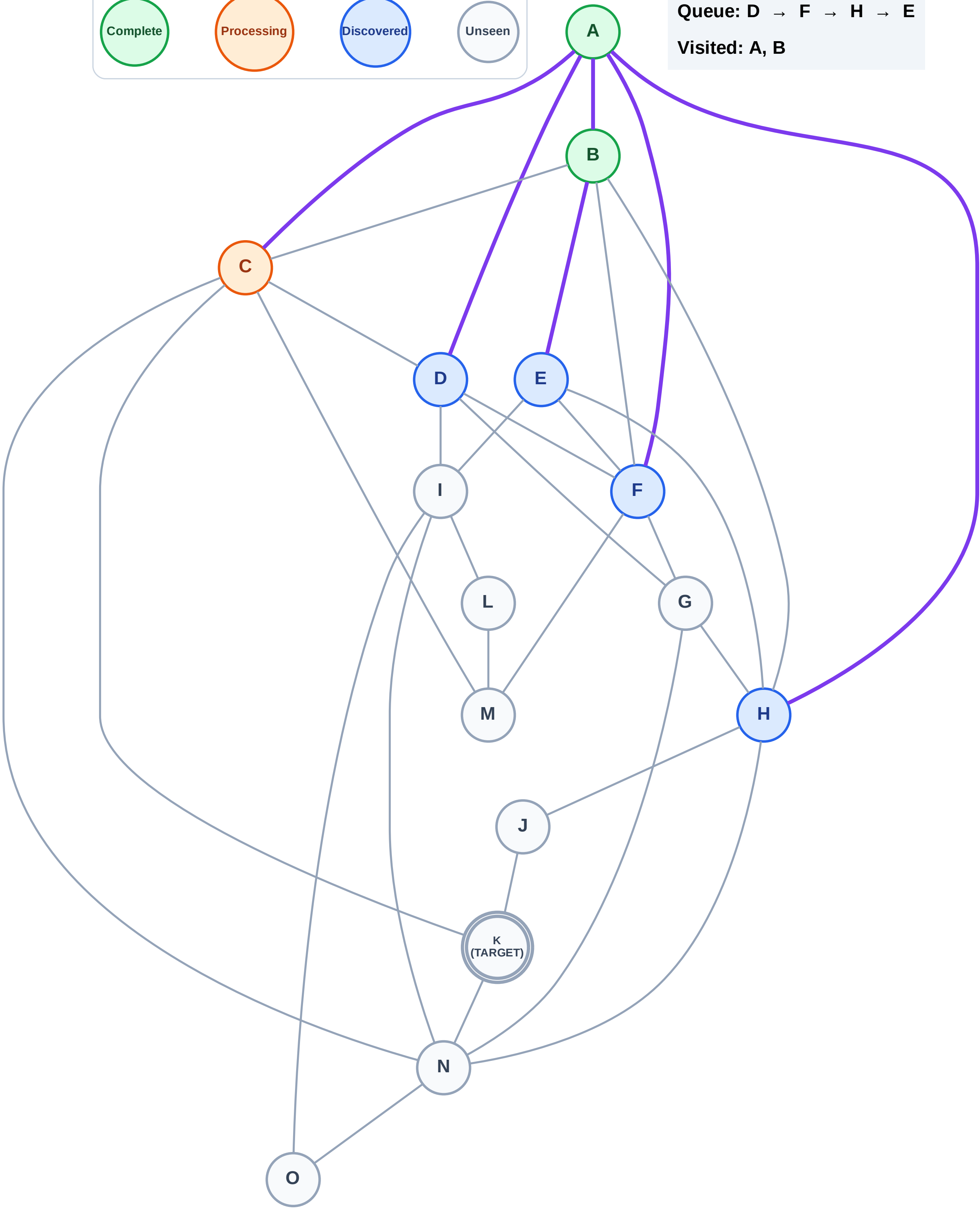
Processing

Discovered

Unseen

Queue: D → F → H → E

Visited: A, B



# BFS Traversal

Step 7: Discover K, M, N from C

Legend

Complete

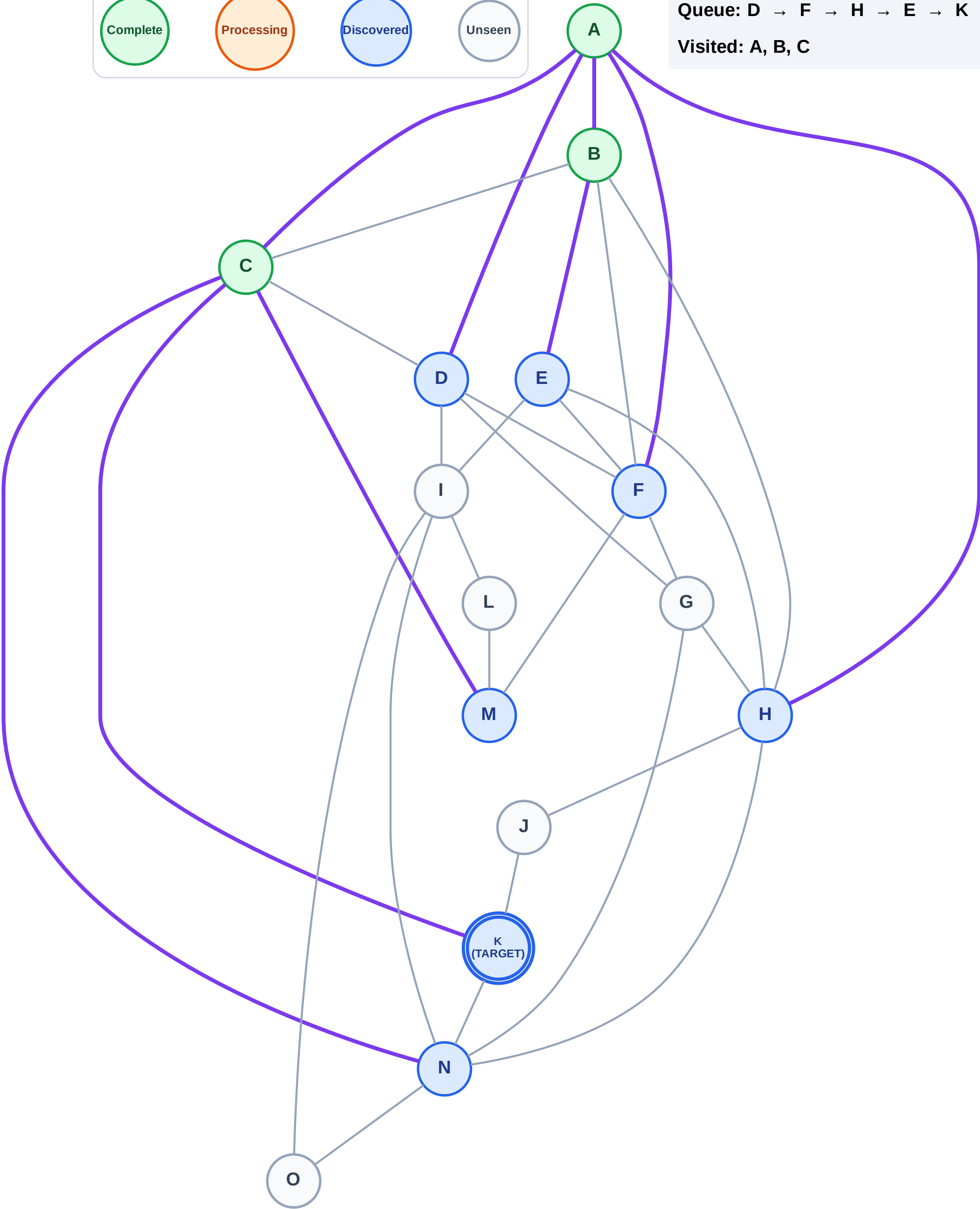
Processing

Discovered

Unseen

Queue: D → F → H → E → K → M → N

Visited: A, B, C



# BFS Traversal

Step 8: Process node D

Legend

Complete

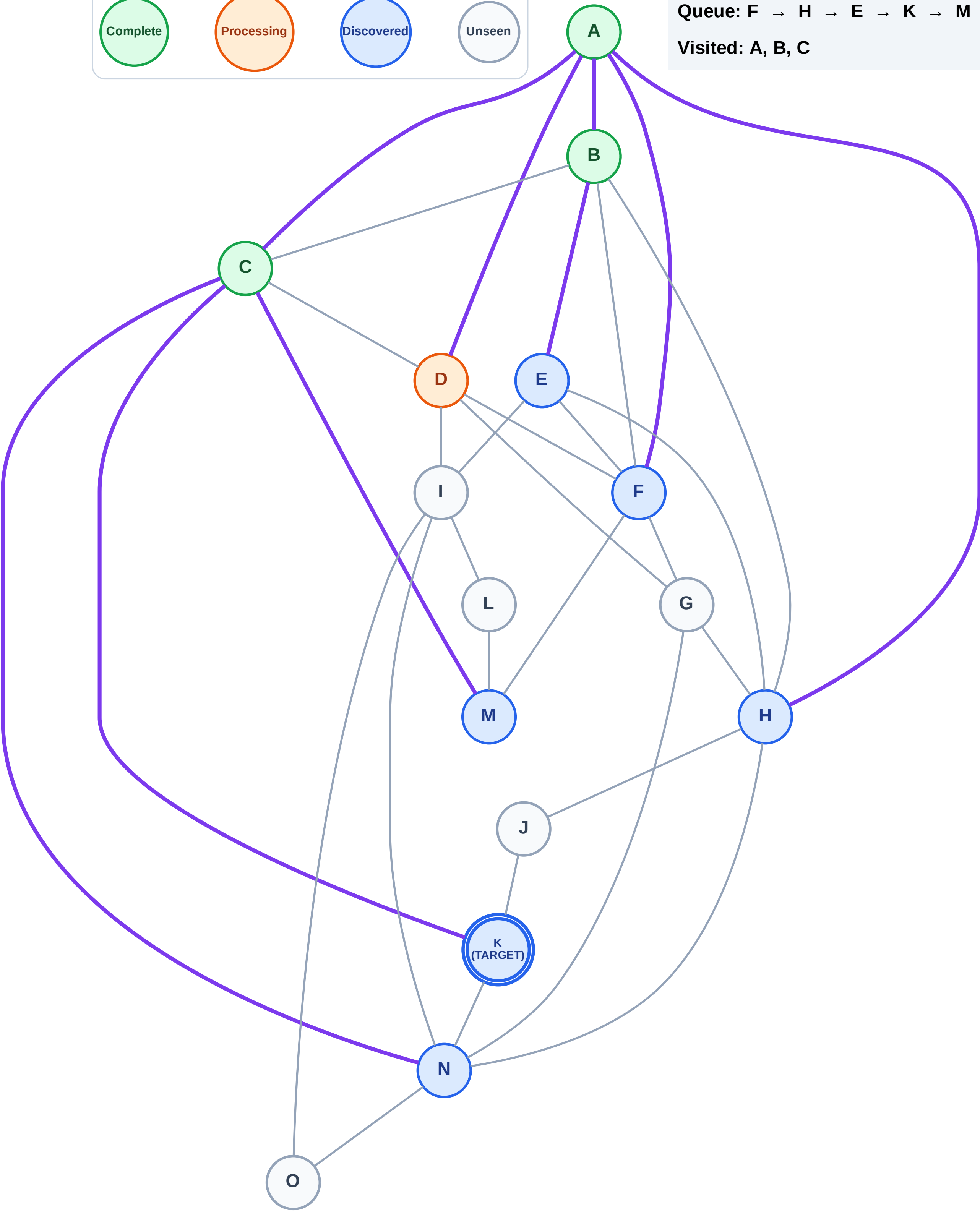
Processing

Discovered

Unseen

Queue: F → H → E → K → M → N

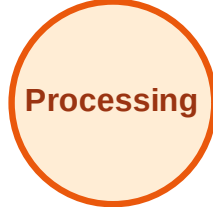
Visited: A, B, C



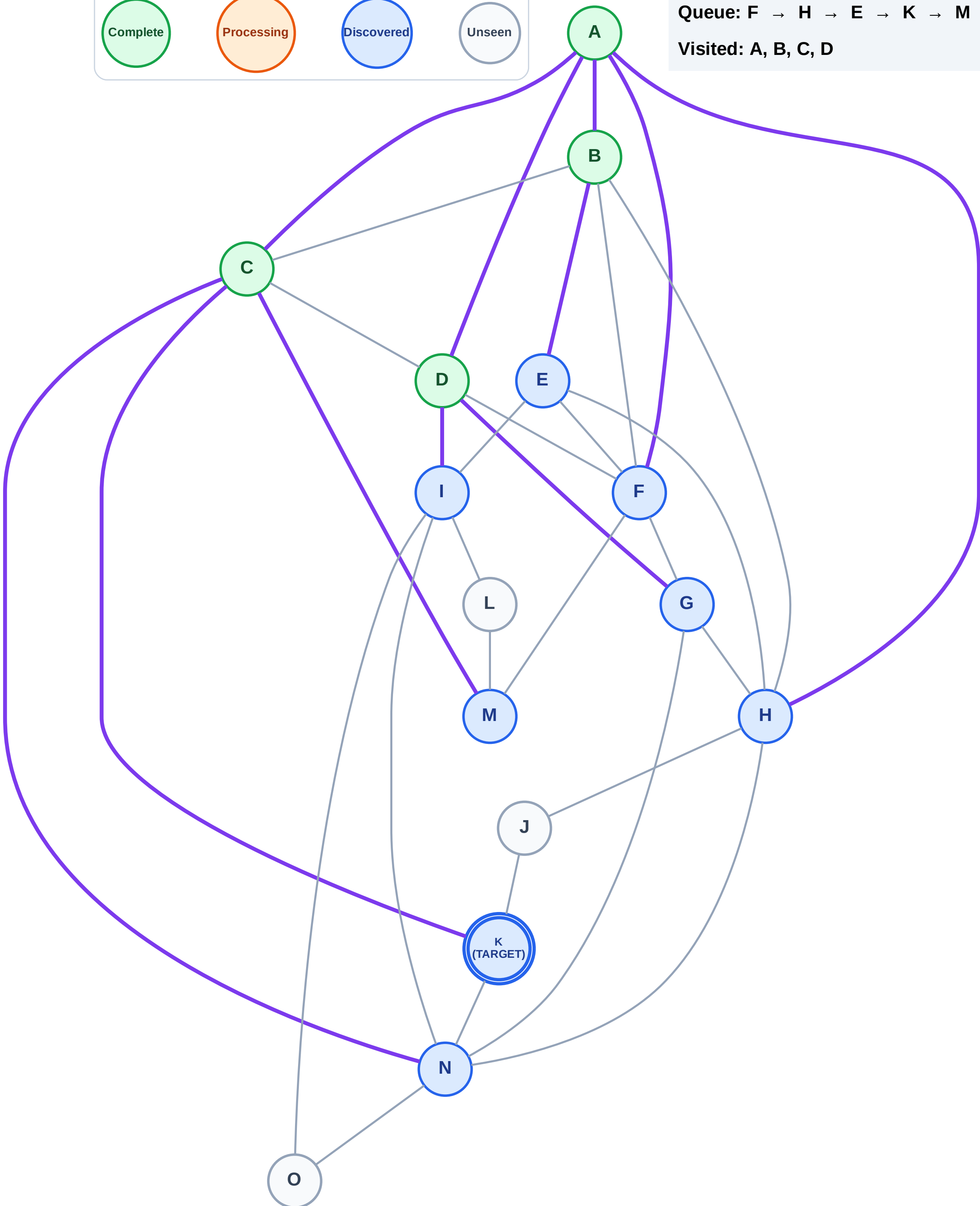
# BFS Traversal

Step 9: Discover G, I from D

## Legend



Queue: F → H → E → K → M → N → G → I  
Visited: A, B, C, D



# BFS Traversal

Step 10: Process node F

Legend

Complete

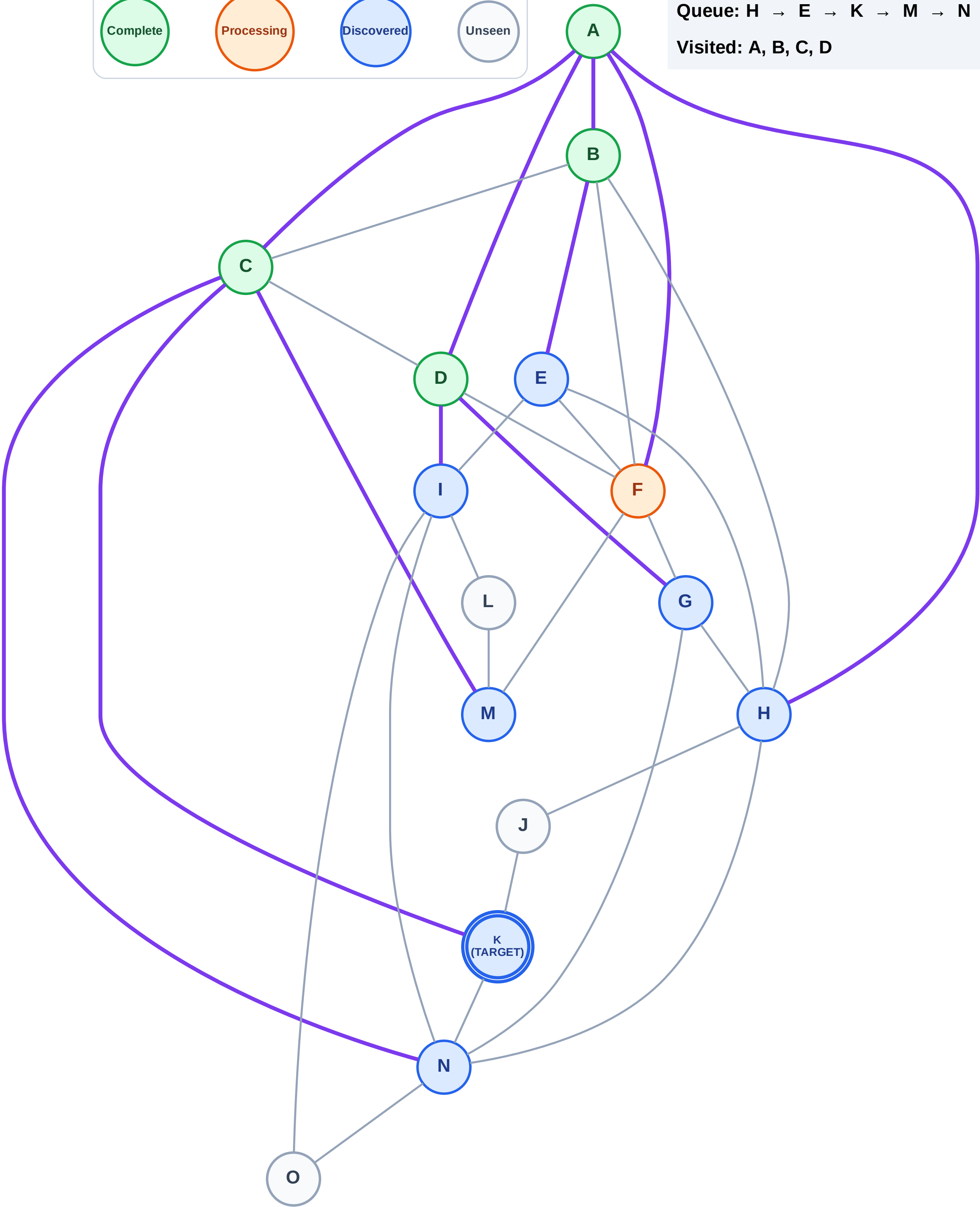
Processing

Discovered

Unseen

Queue: H → E → K → M → N → G → I

Visited: A, B, C, D



# BFS Traversal

Step 11: No new neighbors from F

Legend

Complete

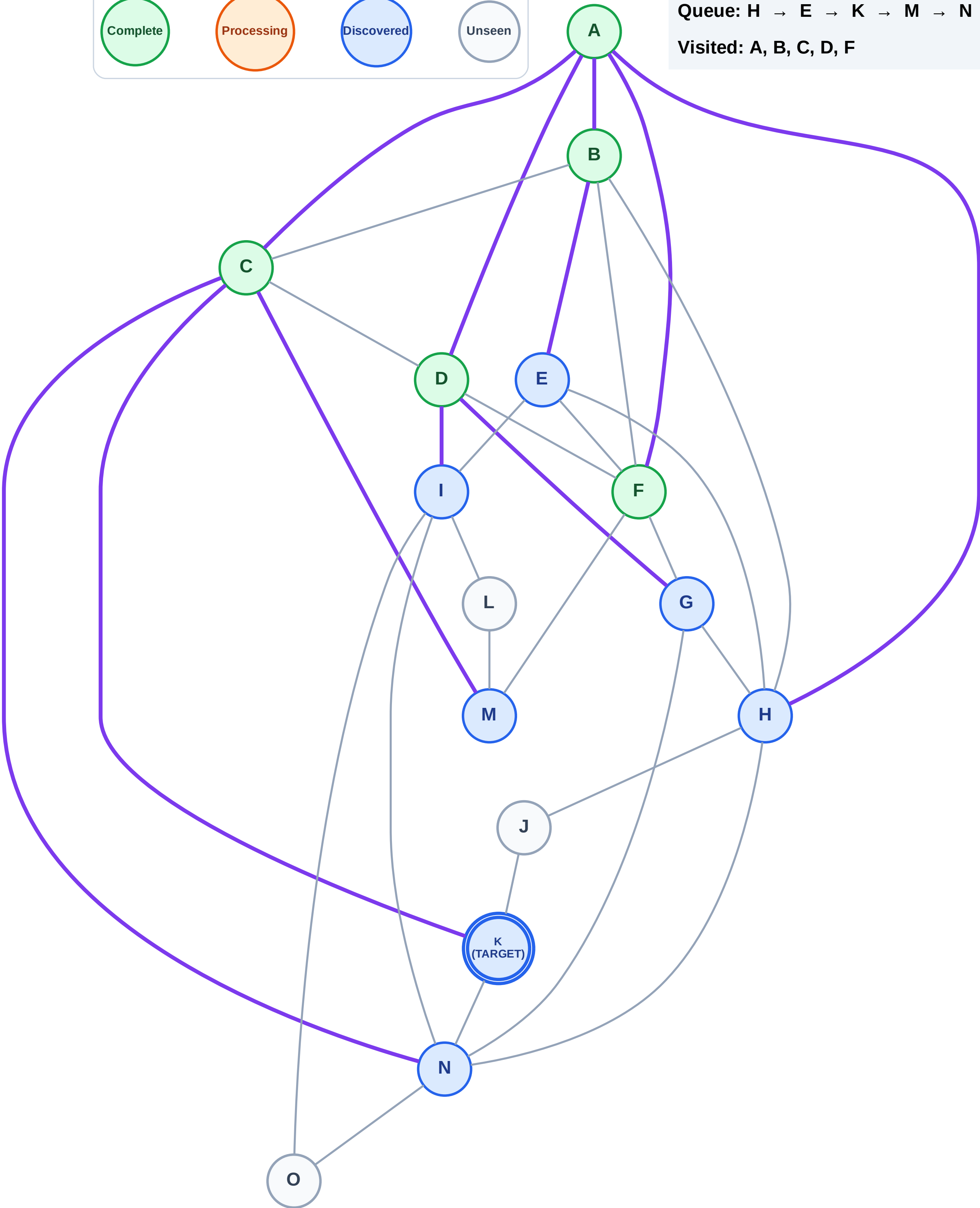
Processing

Discovered

Unseen

Queue: H → E → K → M → N → G → I

Visited: A, B, C, D, F



# BFS Traversal

Step 12: Process node H

Legend

Complete

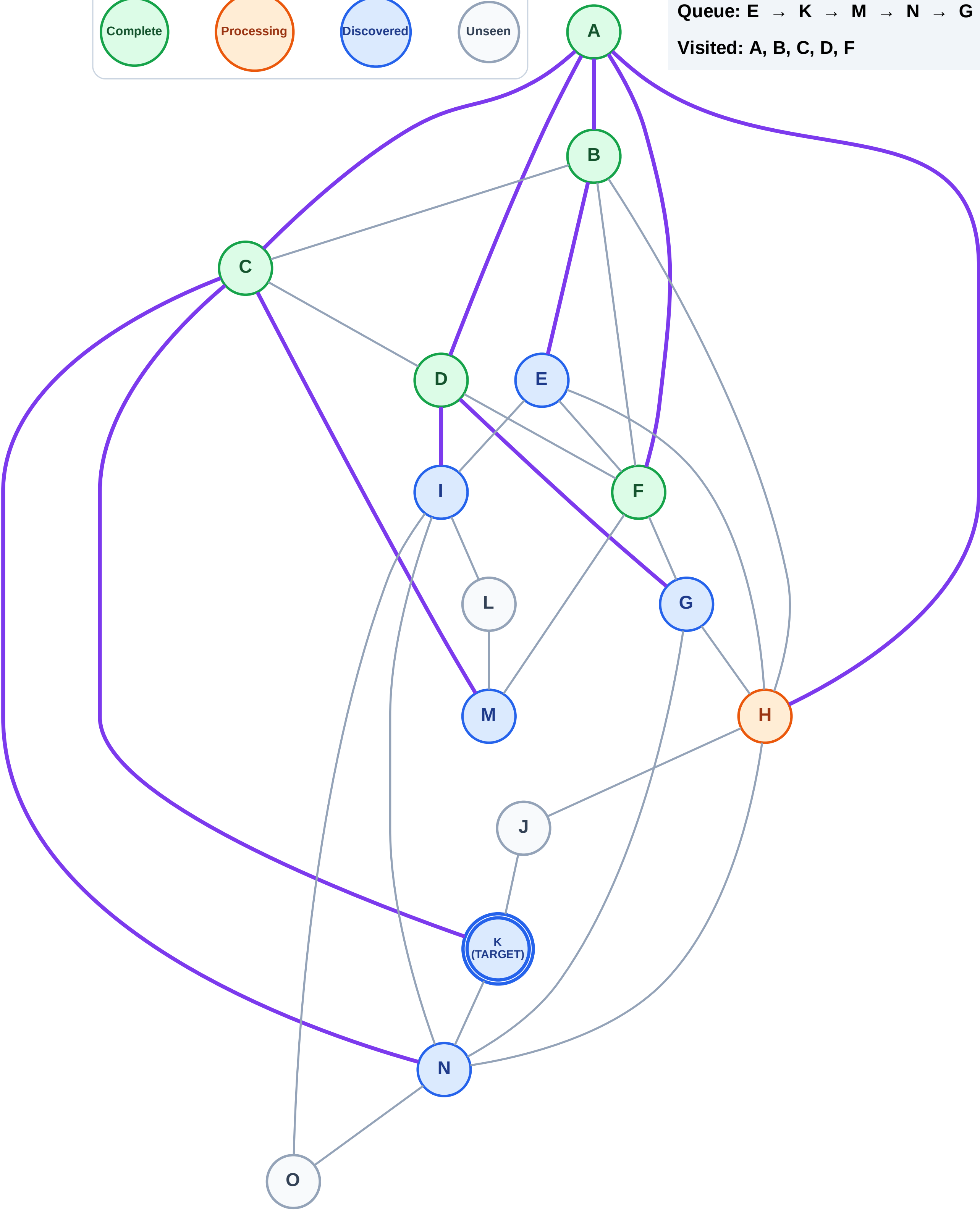
Processing

Discovered

Unseen

Queue: E → K → M → N → G → I

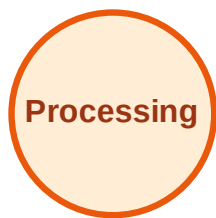
Visited: A, B, C, D, F



# BFS Traversal

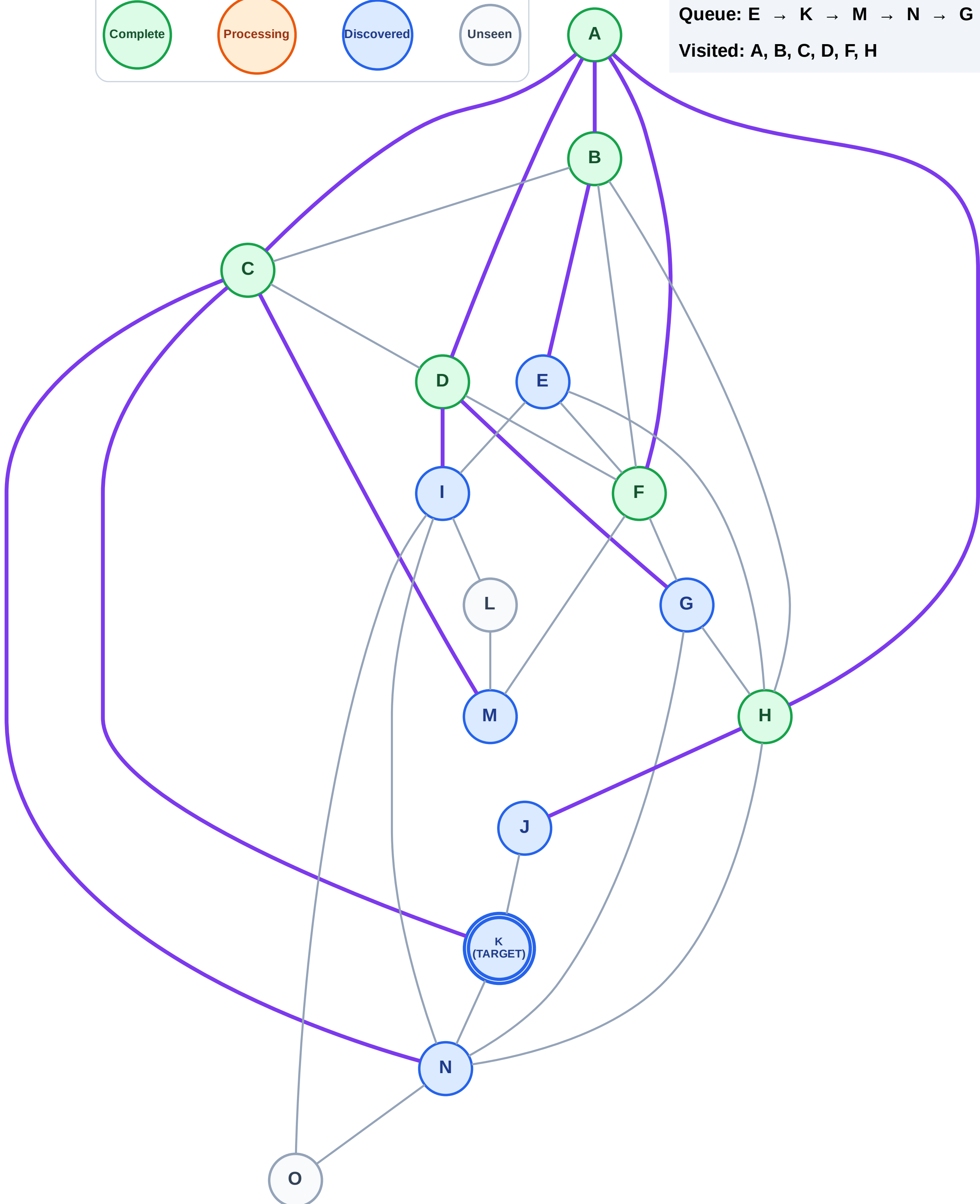
Step 13: Discover J from H

## Legend



Queue: E → K → M → N → G → I → J

Visited: A, B, C, D, F, H



# BFS Traversal

Step 14: Process node E

Legend

Complete

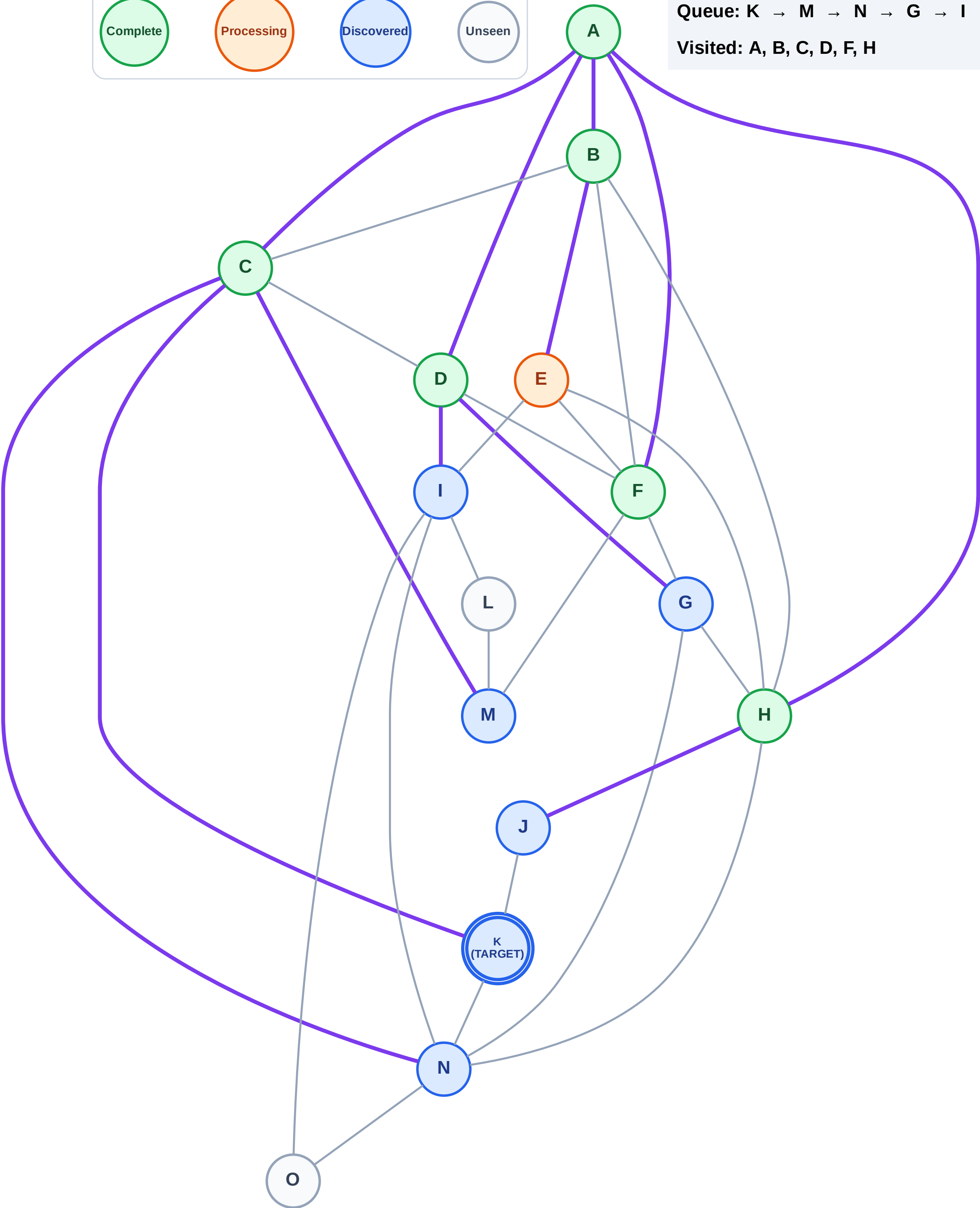
Processing

Discovered

Unseen

Queue: K → M → N → G → I → J

Visited: A, B, C, D, F, H



# BFS Traversal

Step 15: No new neighbors from E

Legend

Complete

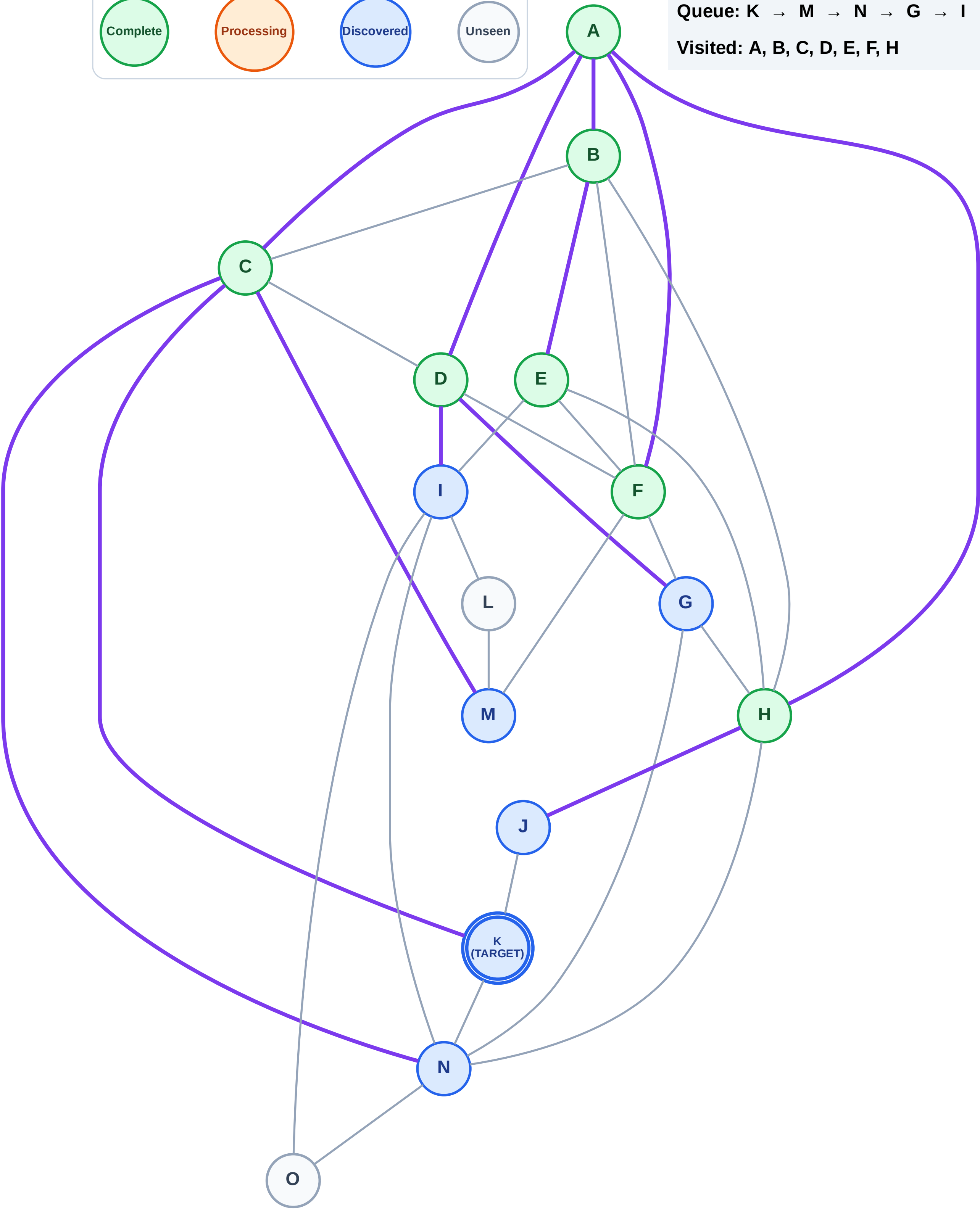
Processing

Discovered

Unseen

Queue: K → M → N → G → I → J

Visited: A, B, C, D, E, F, H



# BFS Traversal

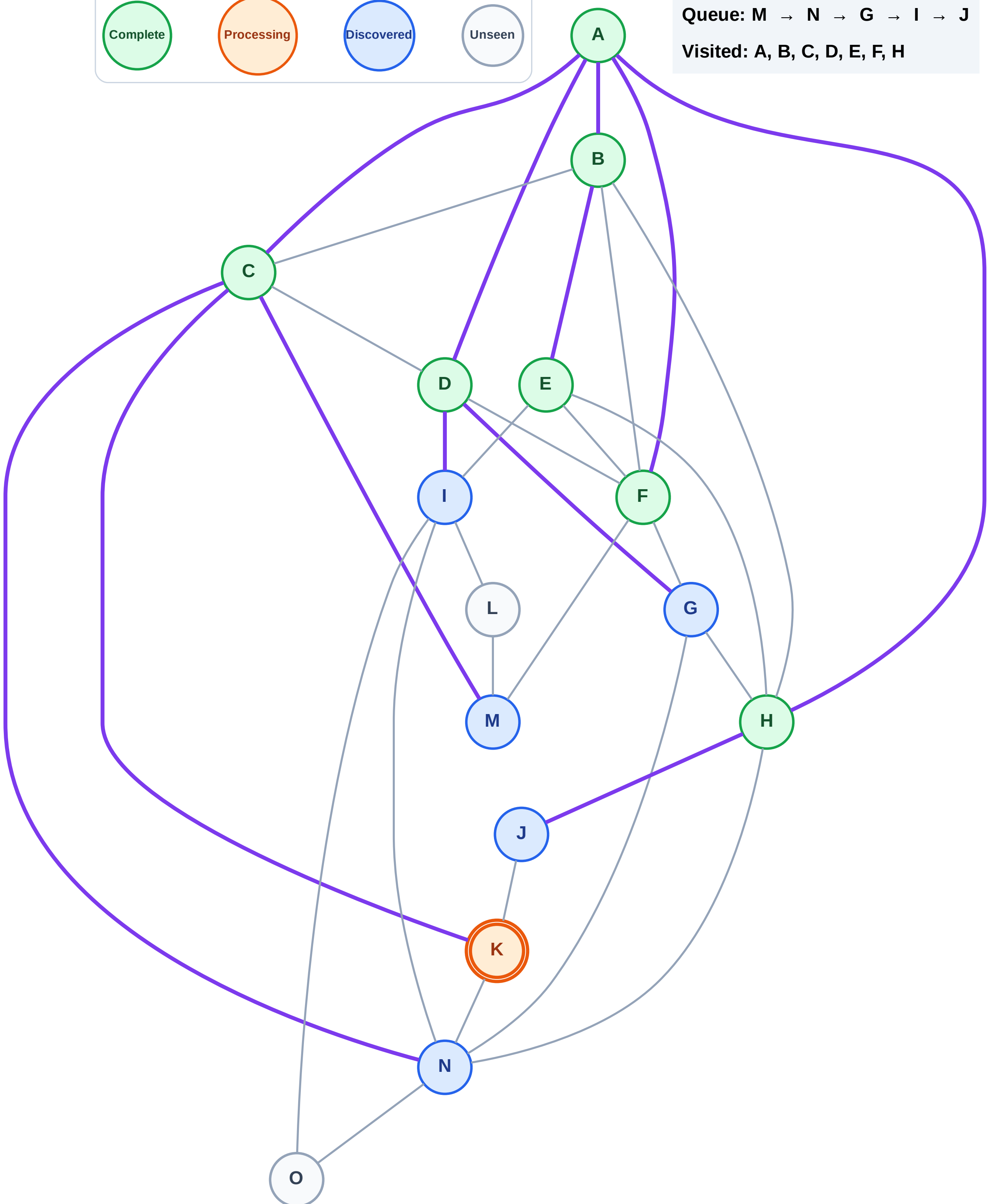
Step 16: Process node K

## Legend



Queue: M → N → G → I → J

Visited: A, B, C, D, E, F, H



# BFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

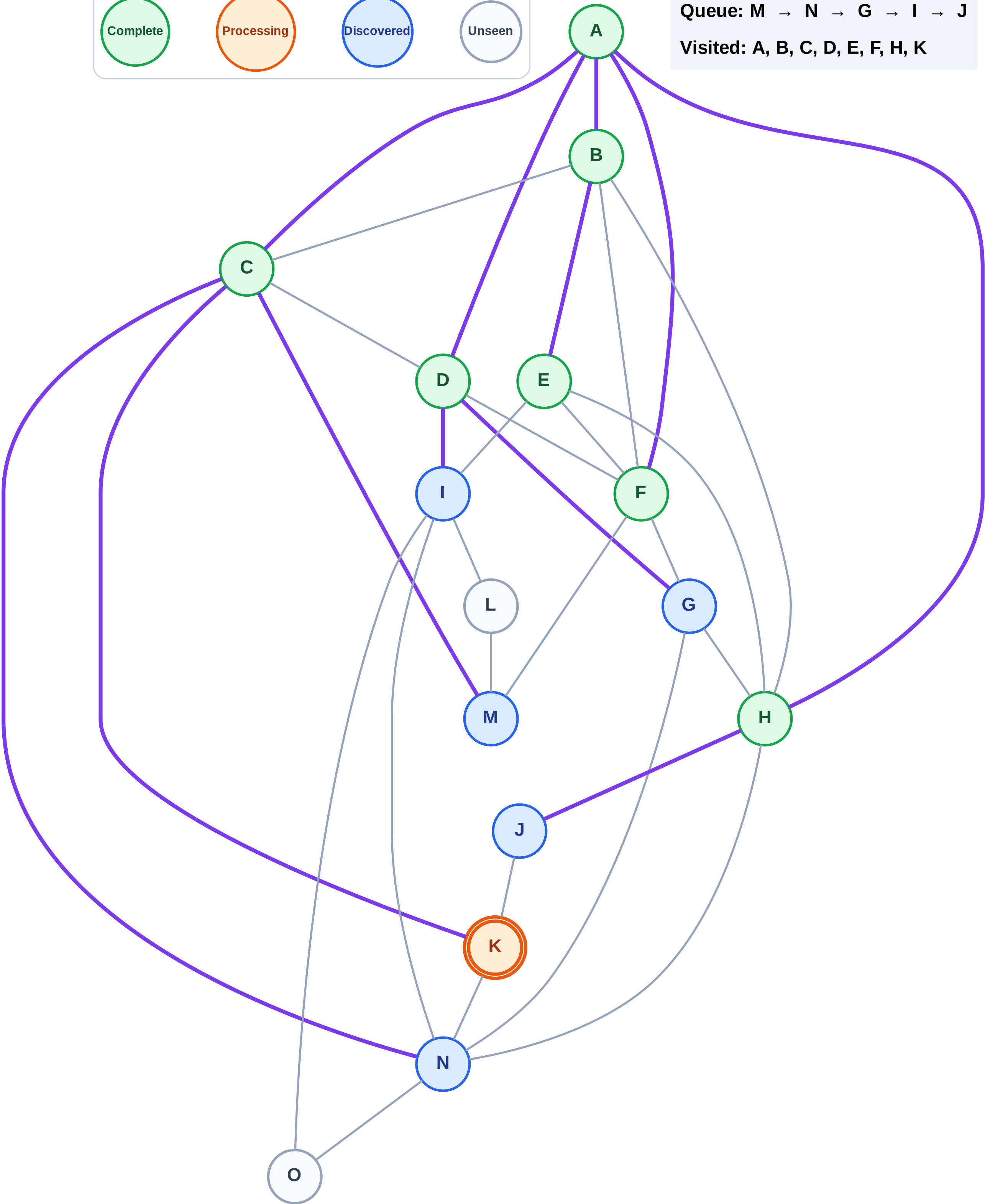
Processing

Discovered

Unseen

Queue: M → N → G → I → J

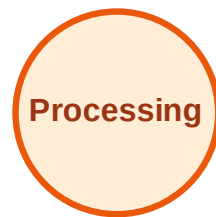
Visited: A, B, C, D, E, F, H, K



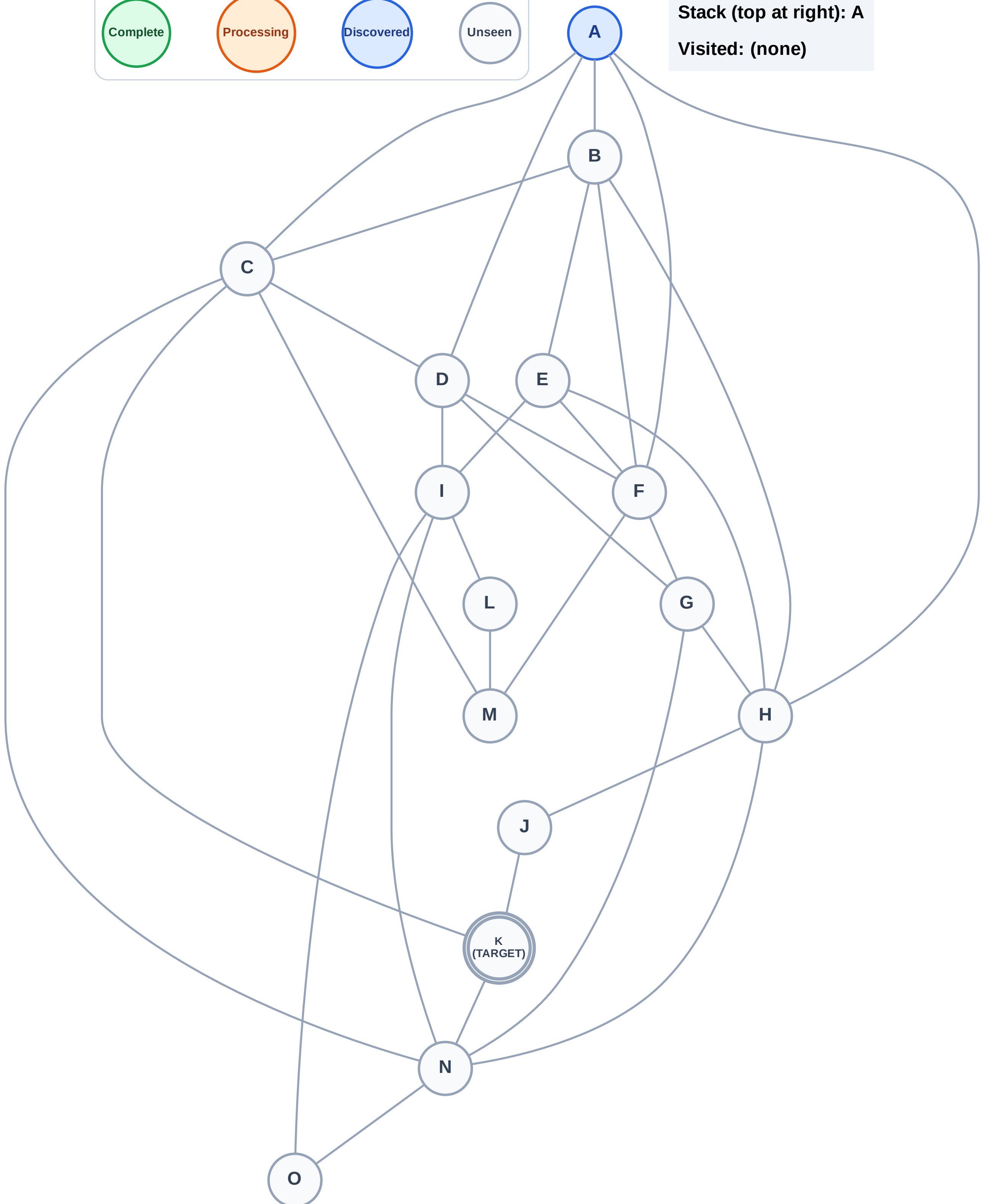
# DFS Traversal

Step 1: Start at A (Seeking target node K)

## Legend



Stack (top at right): A  
Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

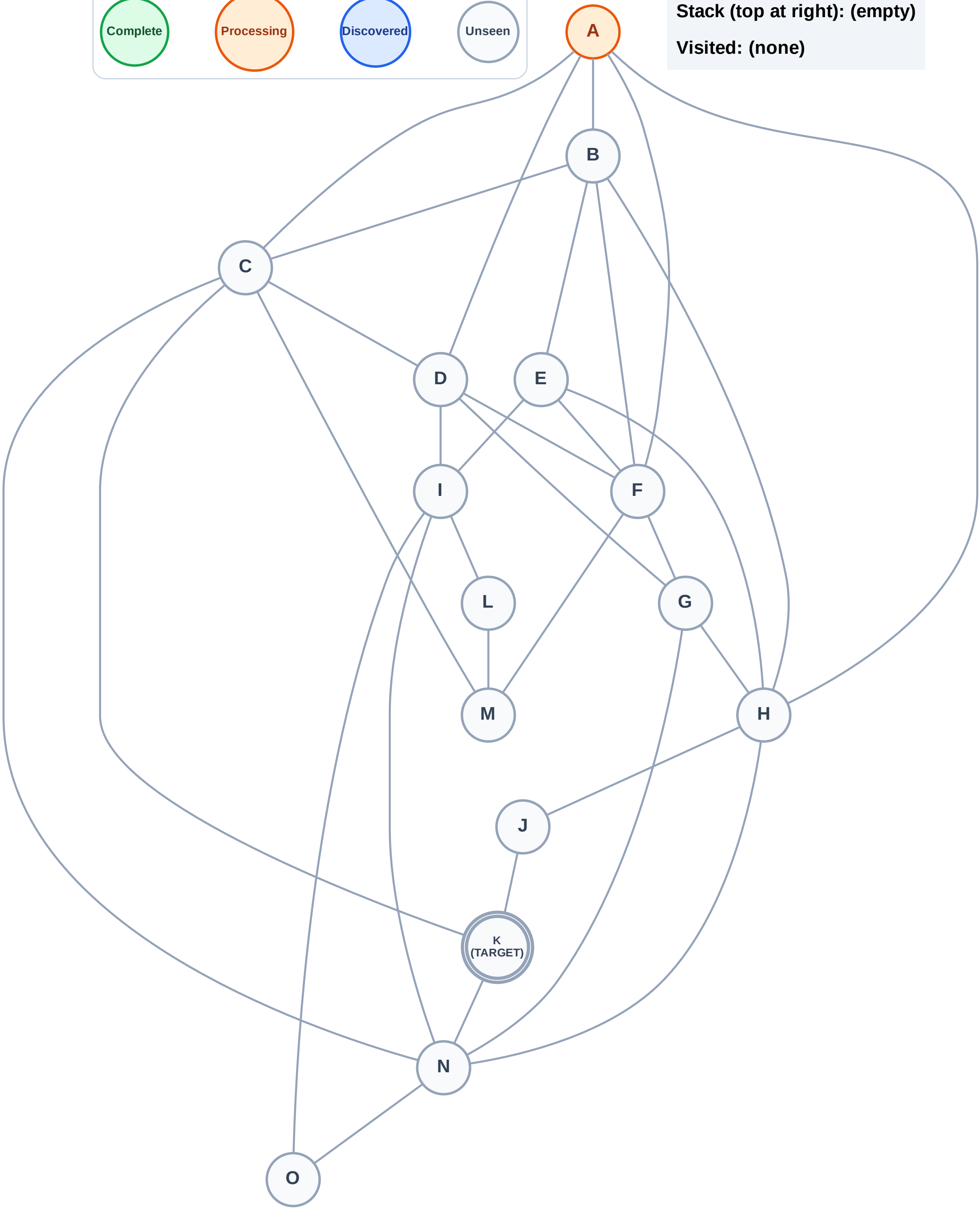
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

Step 3: Discover H, F, D, C, B from A

Legend

Complete

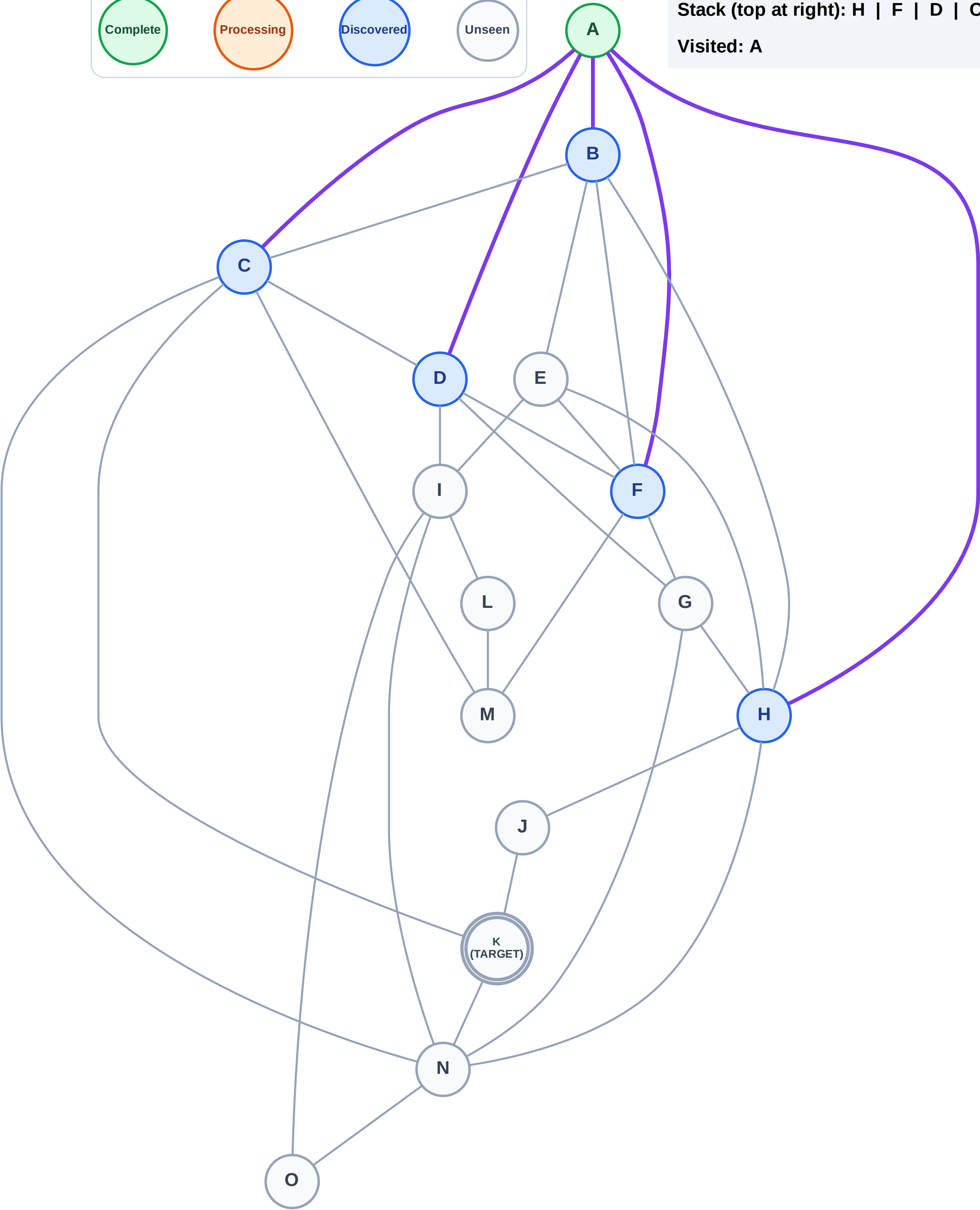
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

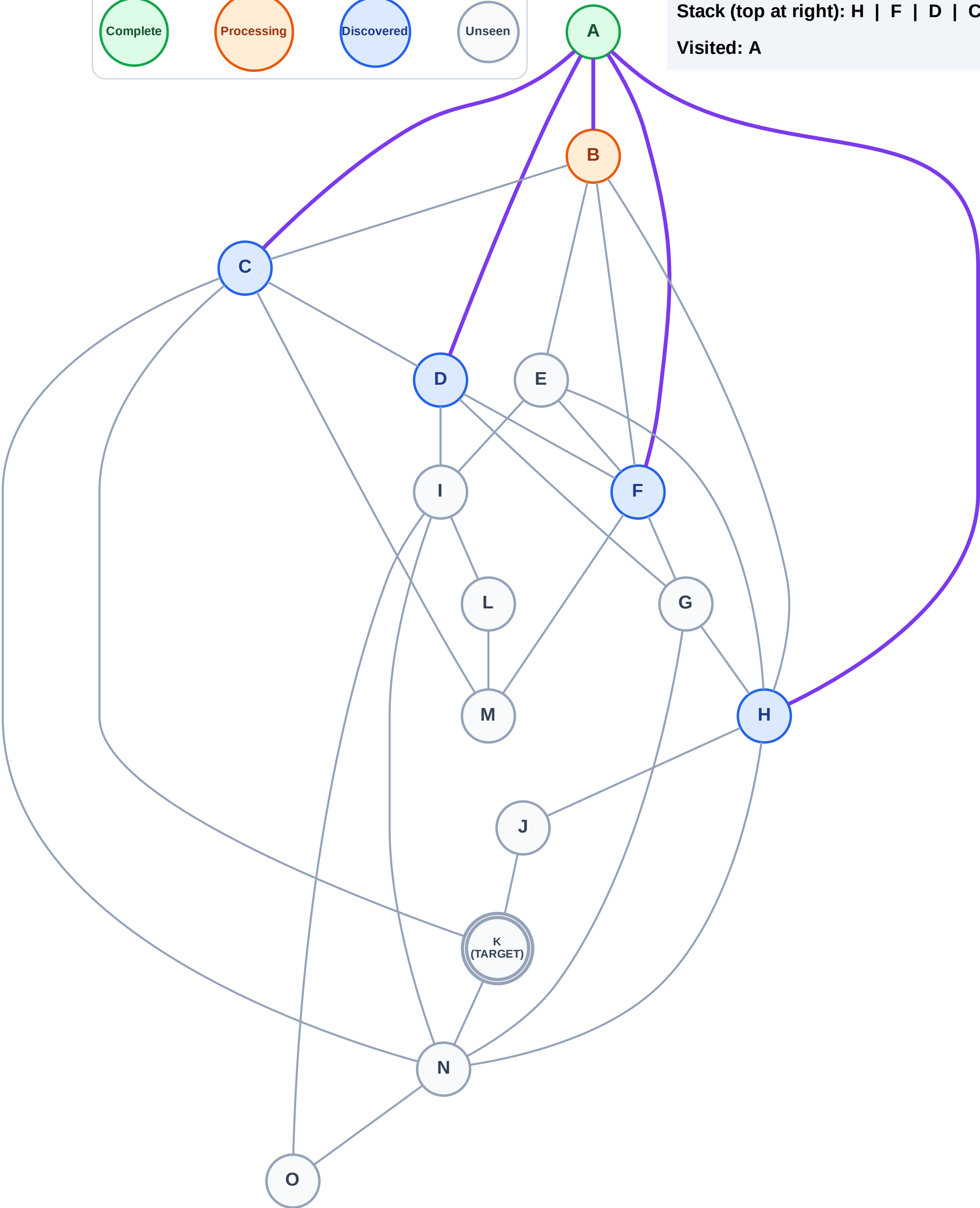
Processing

Discovered

Unseen

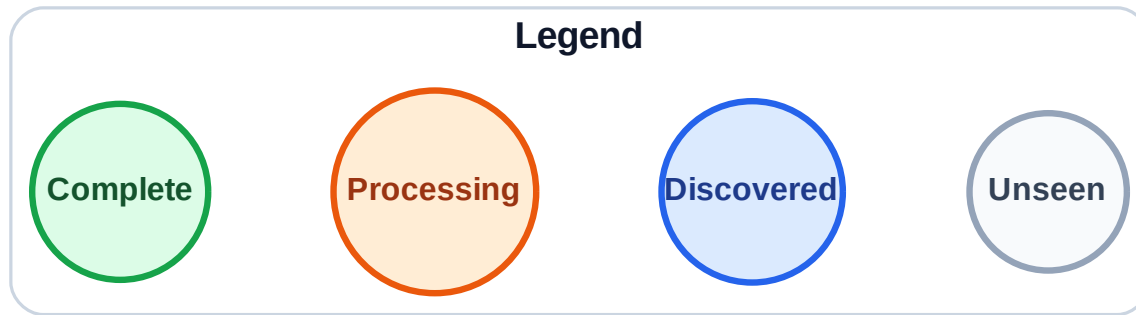
Stack (top at right): H | F | D | C

Visited: A



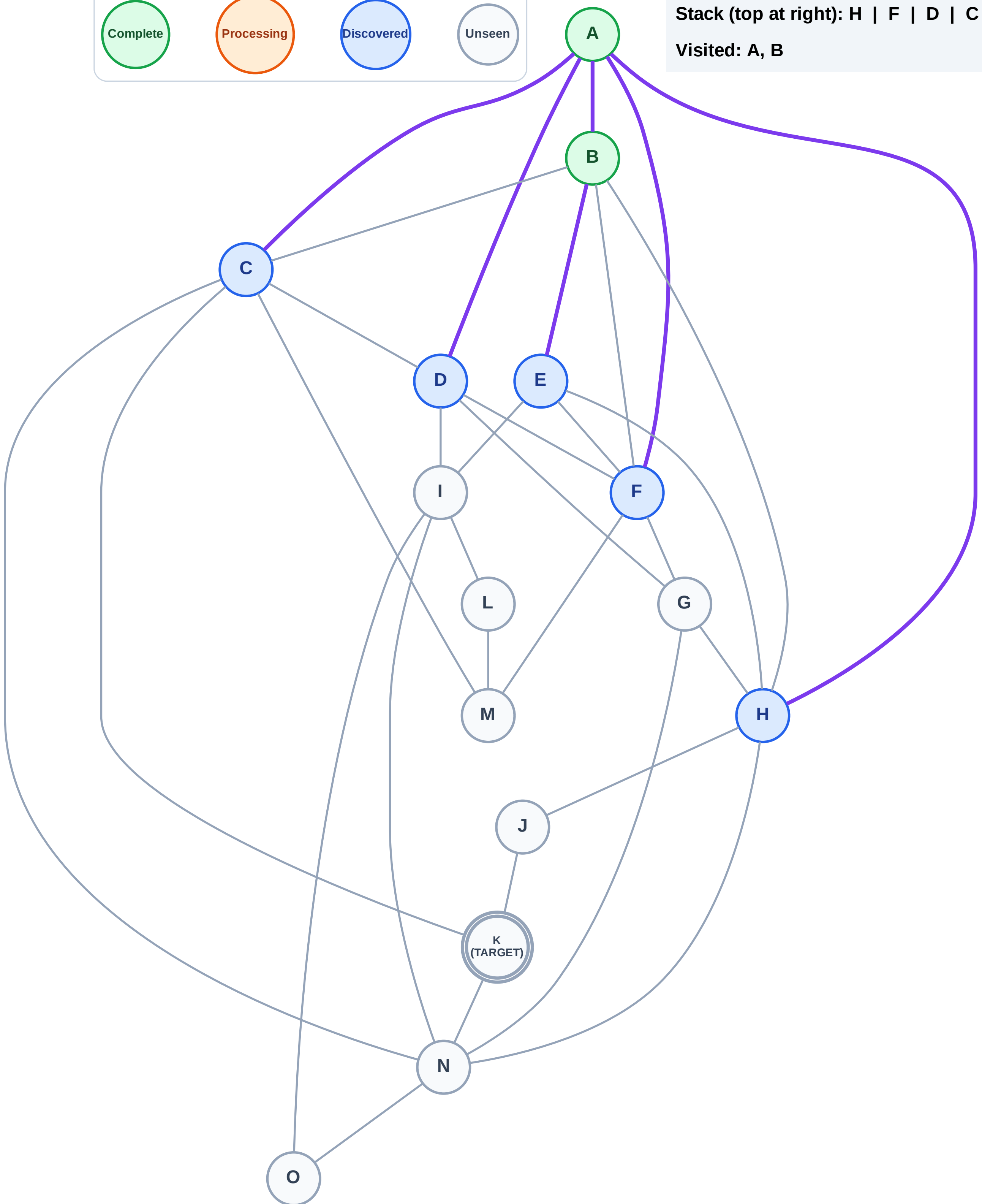
# DFS Traversal

Step 5: Discover E from B



Stack (top at right): H | F | D | C | E

Visited: A, B



# DFS Traversal

Step 6: Process node E

Legend

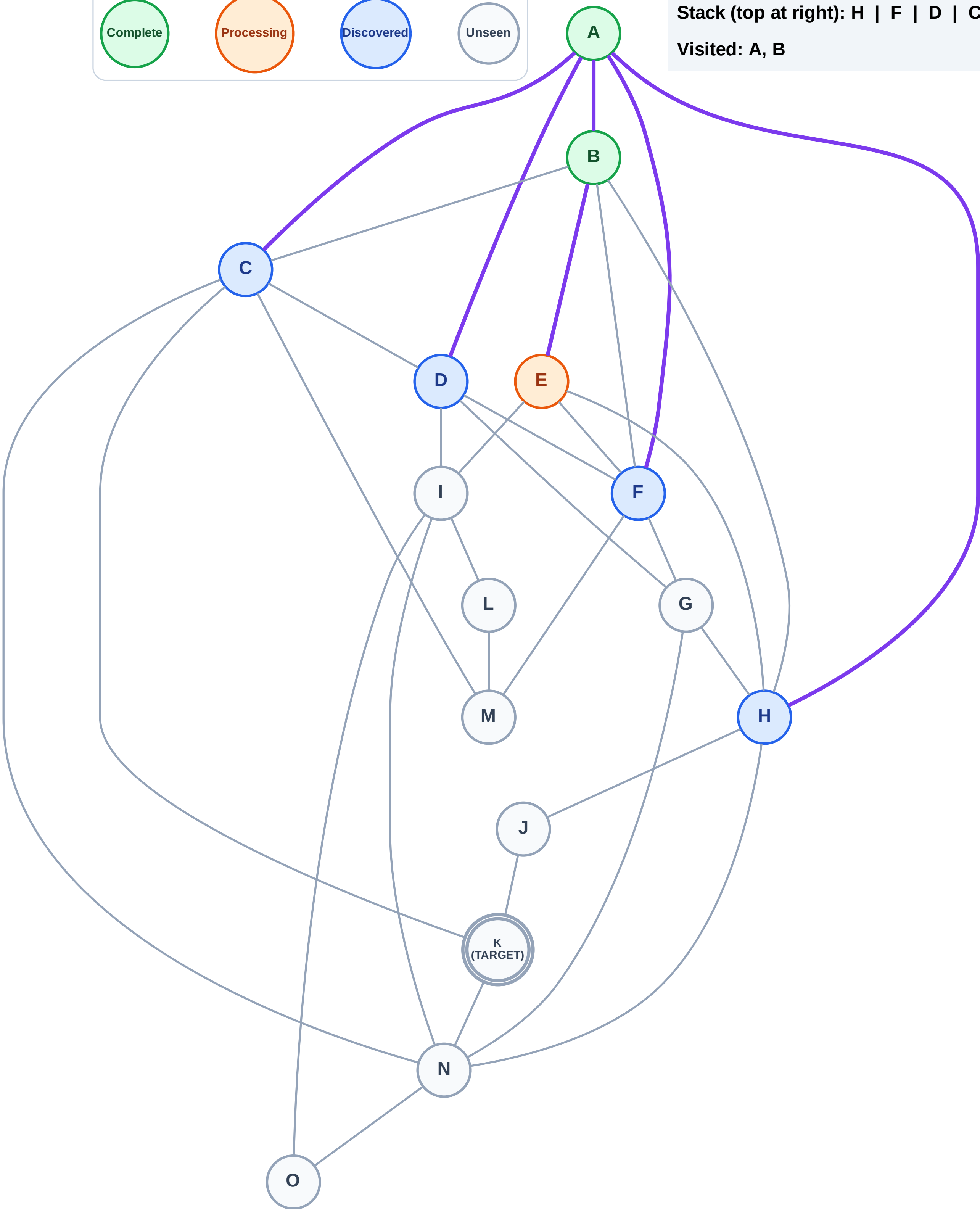
Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | D | C  
Visited: A, B



# DFS Traversal

Step 7: Discover I from E

Legend

Complete

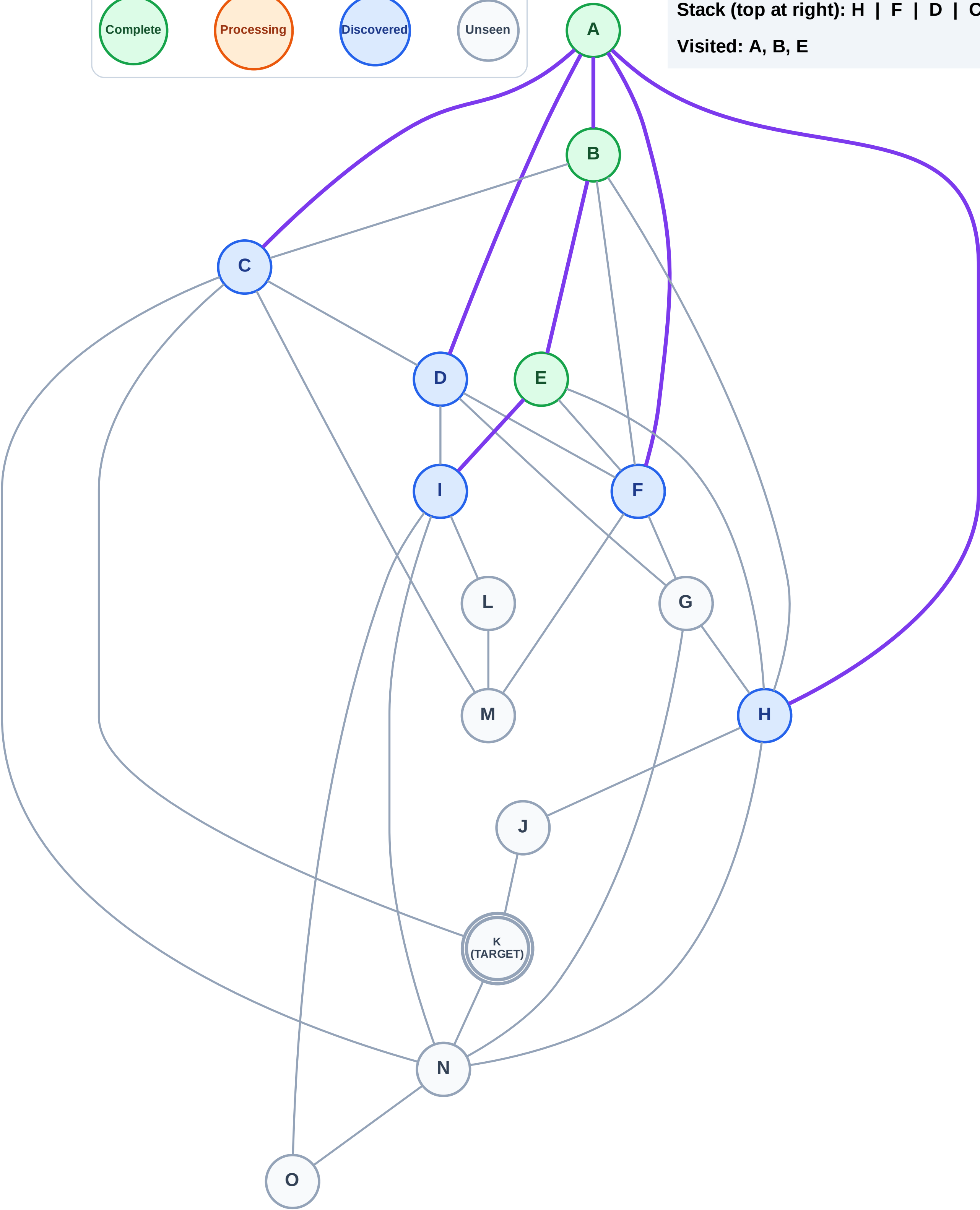
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | I

Visited: A, B, E



# DFS Traversal

Step 8: Process node I

Legend

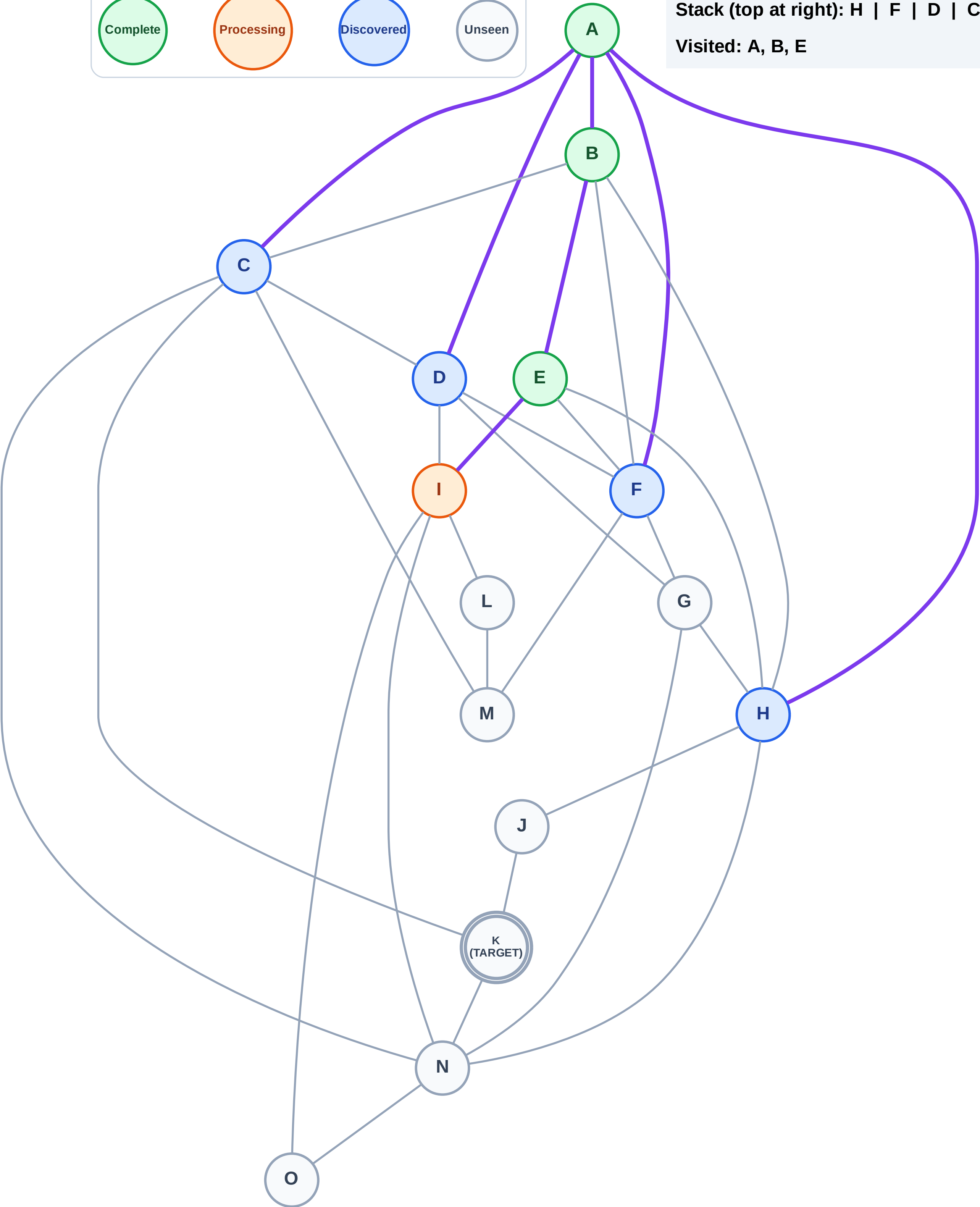
Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | D | C  
Visited: A, B, E



# DFS Traversal

Step 9: Discover O, N, L from I

Legend

Complete

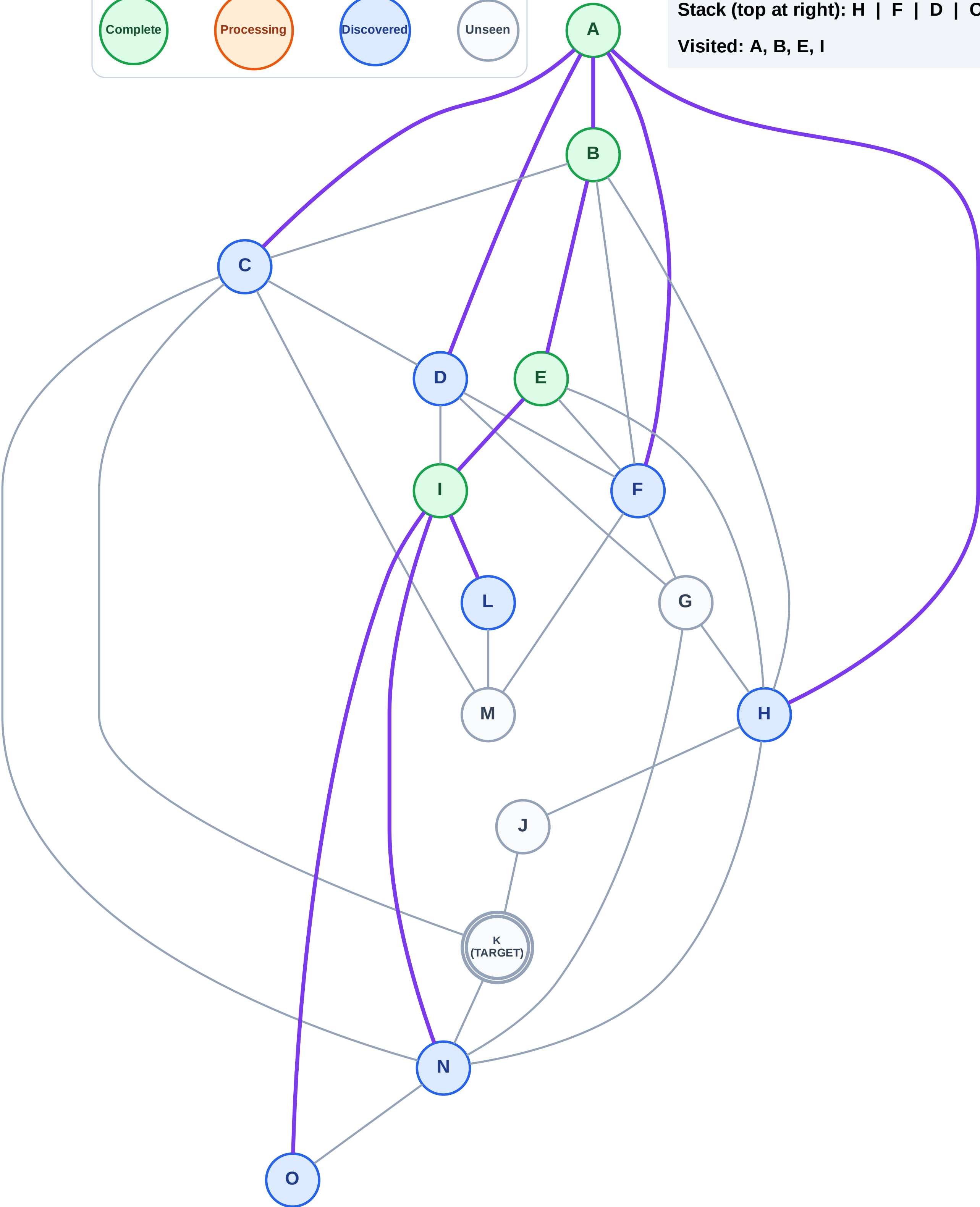
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N | L

Visited: A, B, E, I



# DFS Traversal

Step 10: Process node L

Legend

Complete

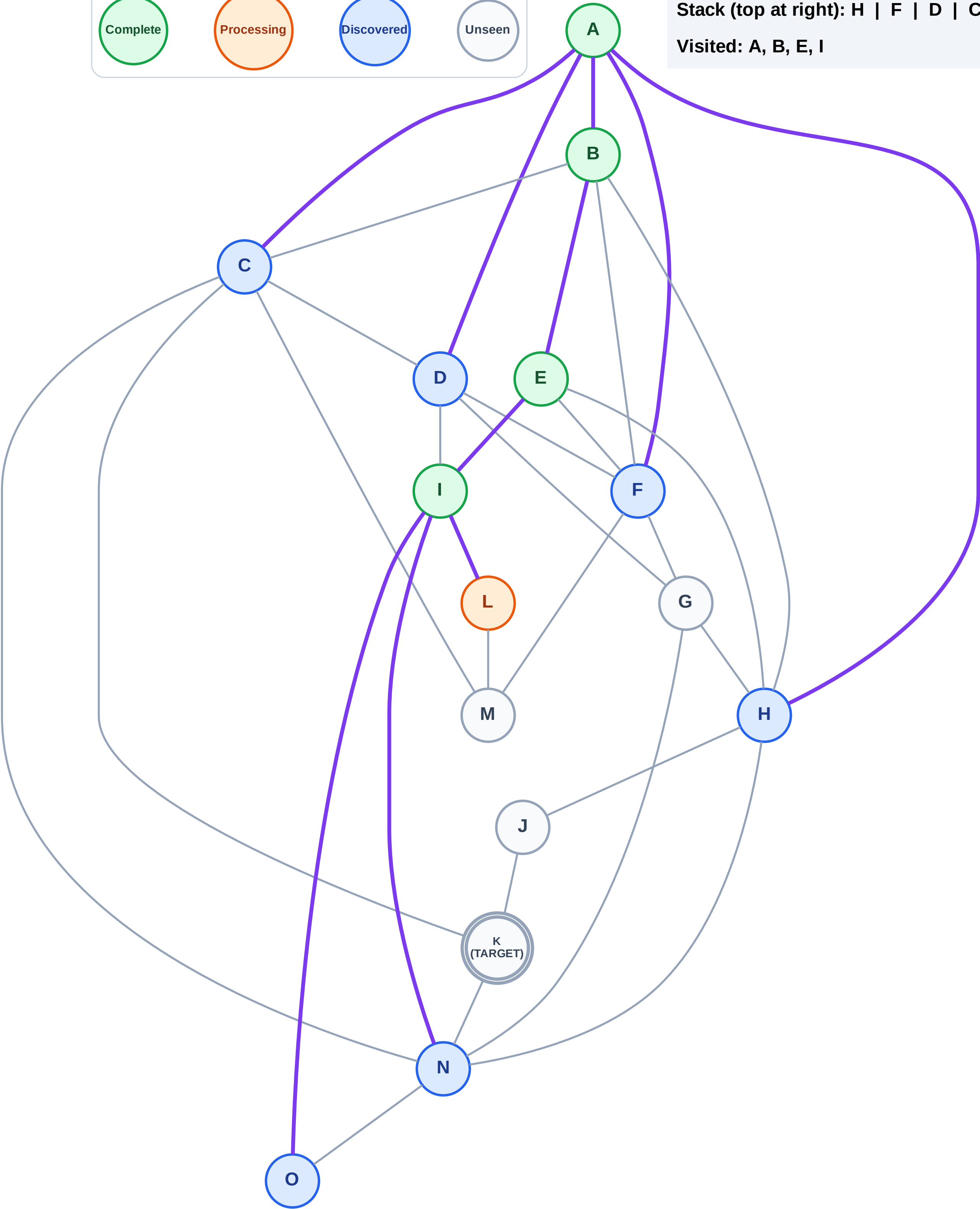
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N

Visited: A, B, E, I



# DFS Traversal

Step 11: Discover M from L

Legend

Complete

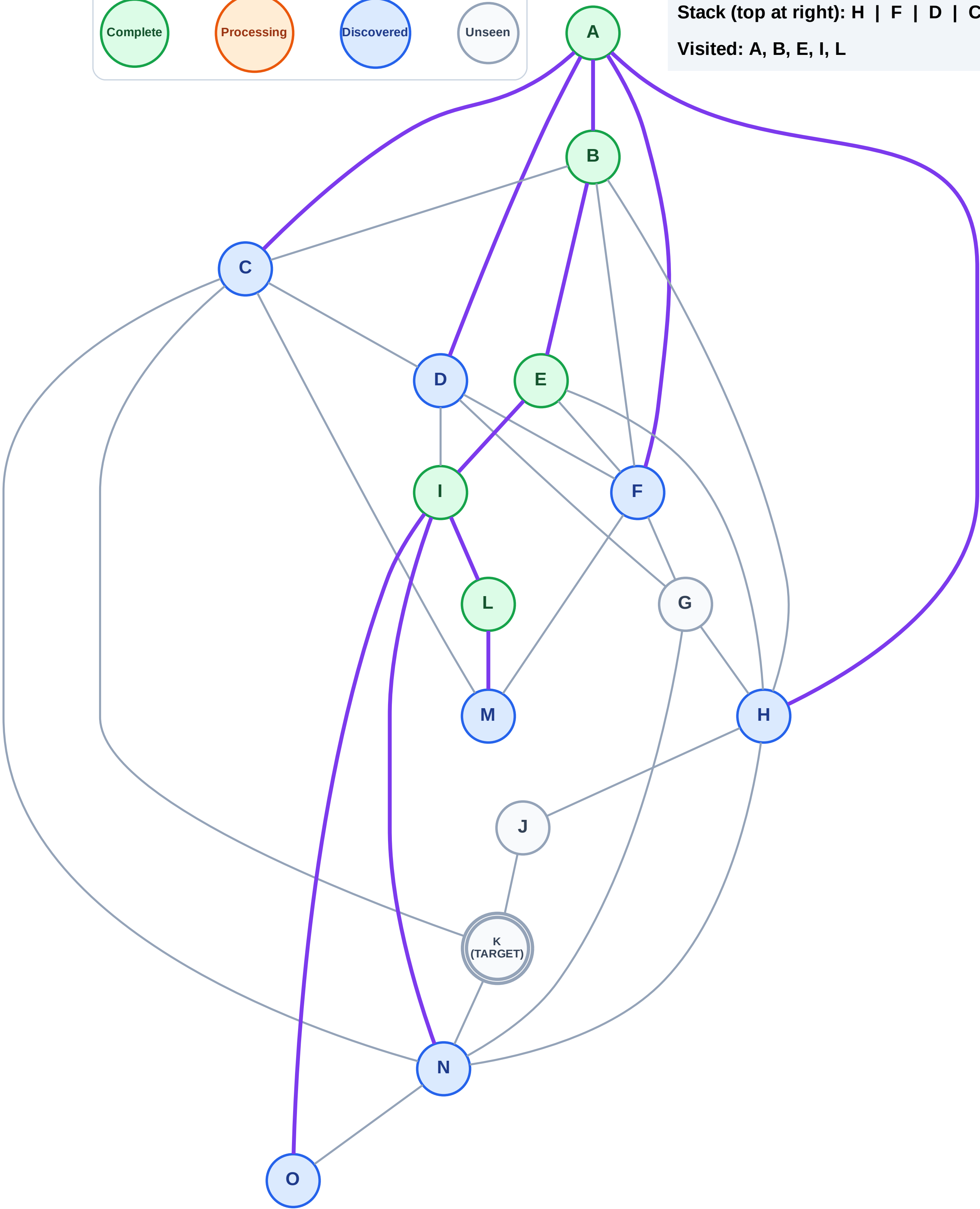
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N | M

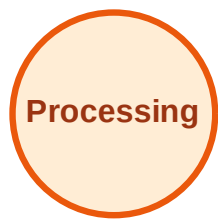
Visited: A, B, E, I, L



# DFS Traversal

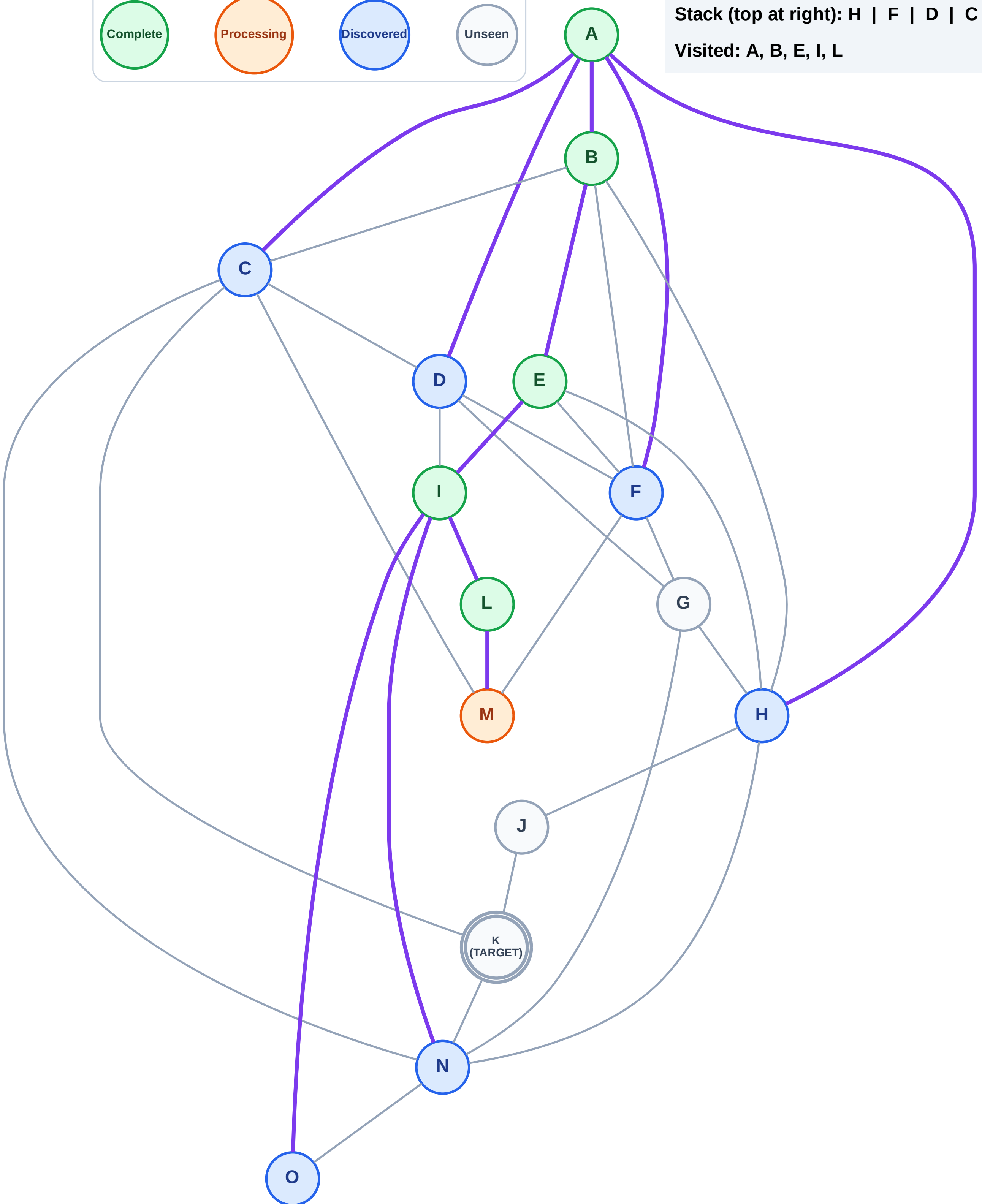
Step 12: Process node M

## Legend



Stack (top at right): H | F | D | C | O | N

Visited: A, B, E, I, L



# DFS Traversal

Step 13: No new neighbors from M

Legend

Complete

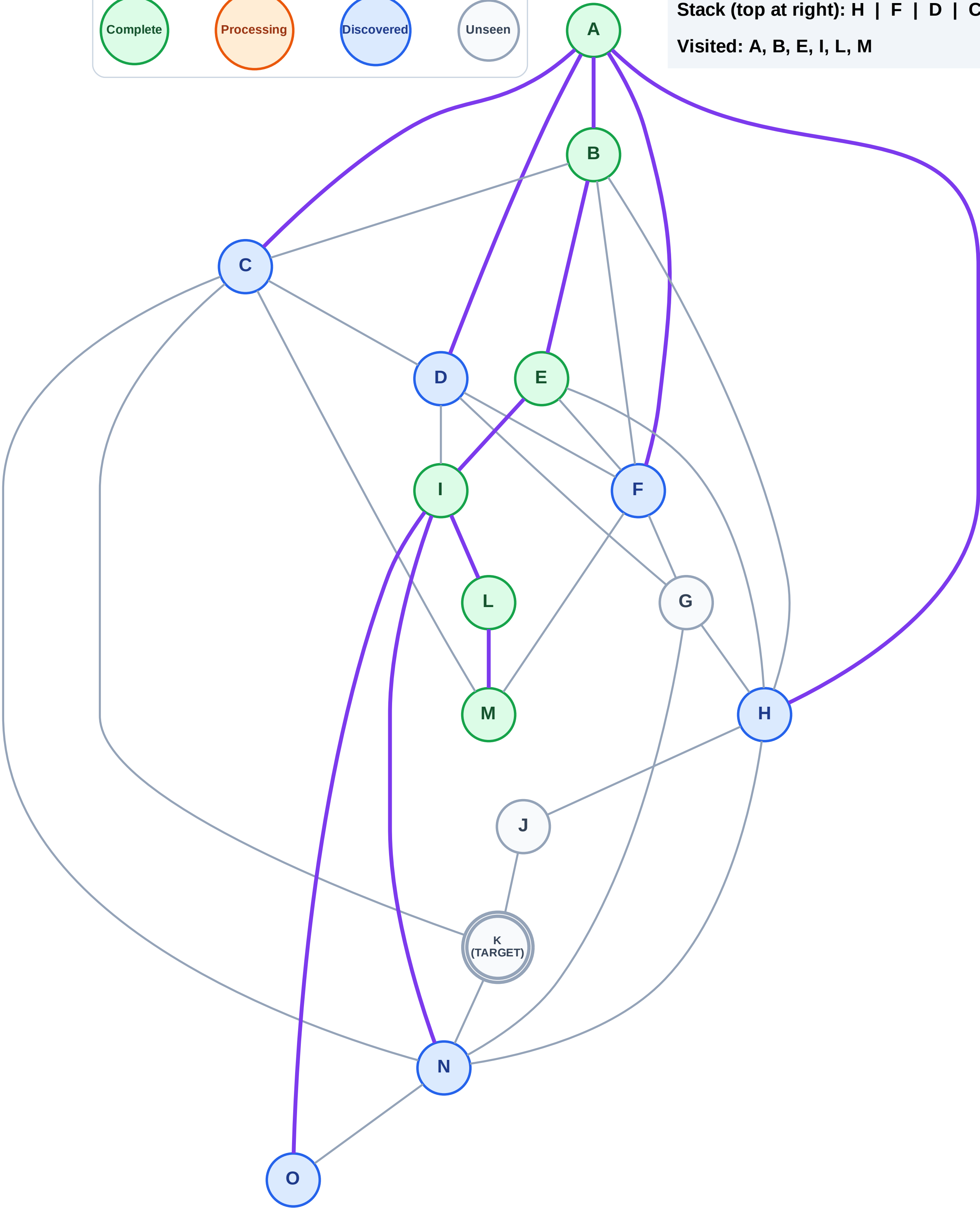
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N

Visited: A, B, E, I, L, M



# DFS Traversal

Step 14: Process node N

Legend

Complete

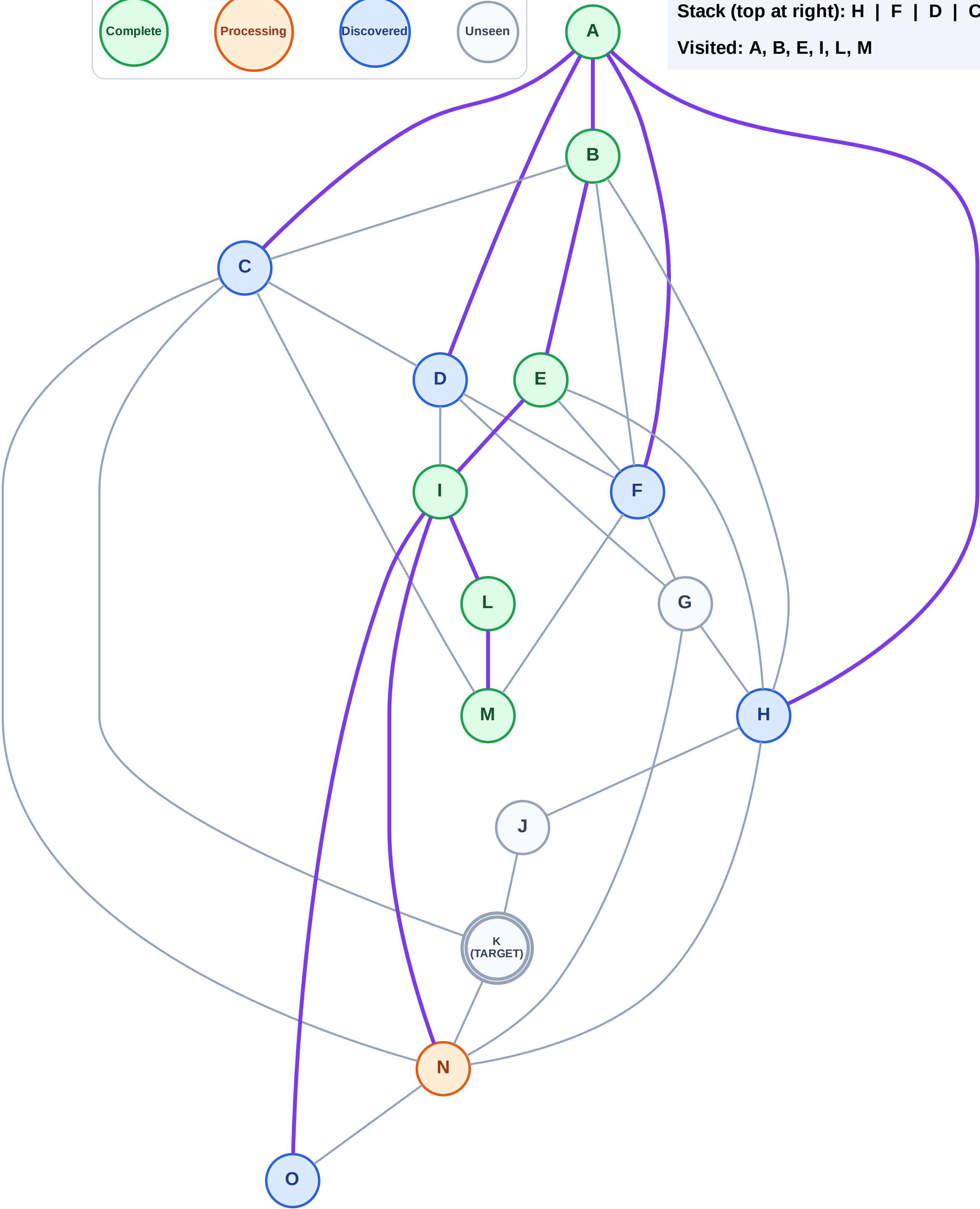
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O

Visited: A, B, E, I, L, M



# DFS Traversal

Step 15: Discover K, G from N

Legend

Complete

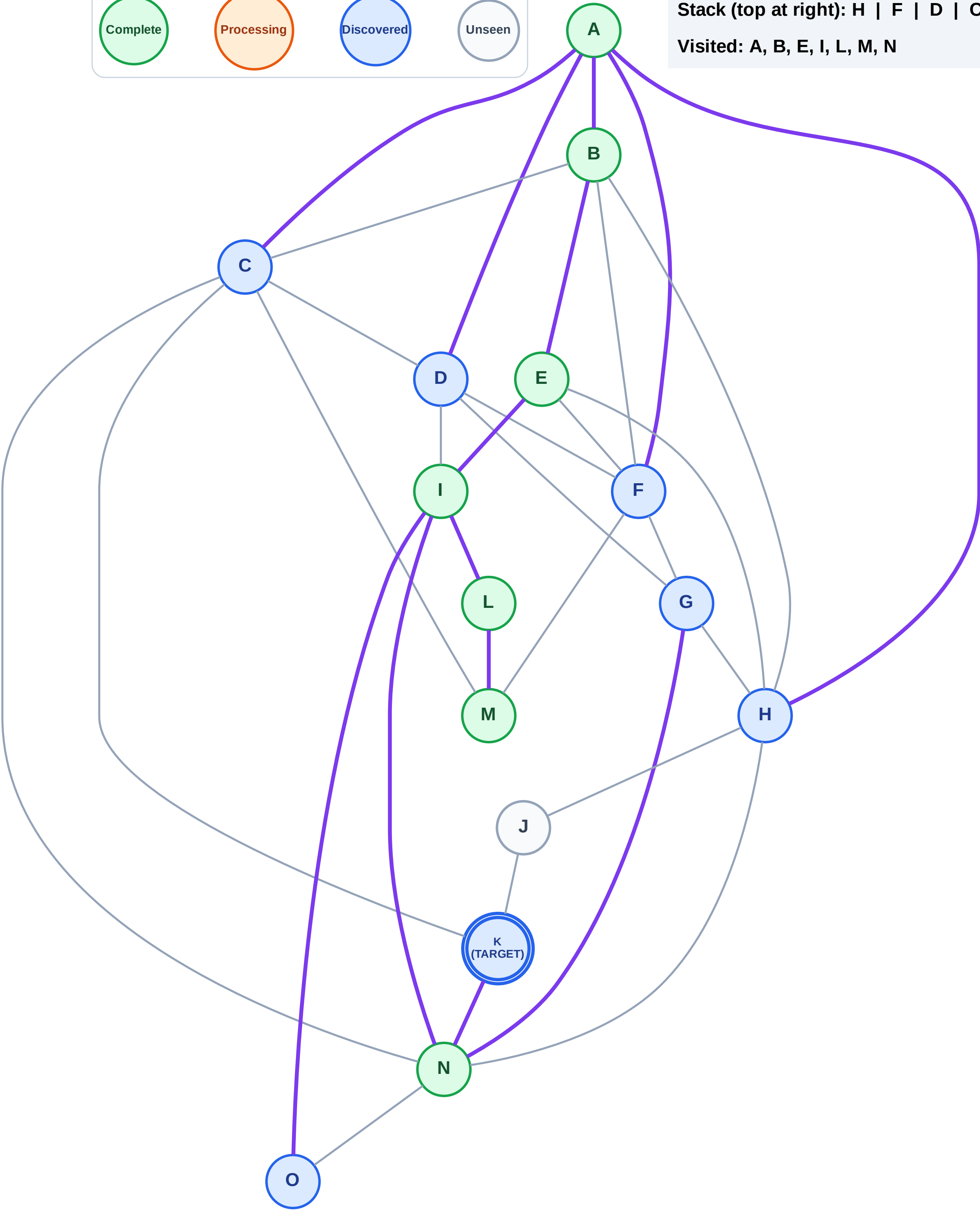
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | K | G

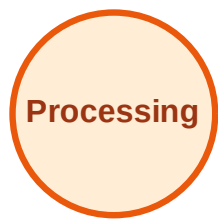
Visited: A, B, E, I, L, M, N



# DFS Traversal

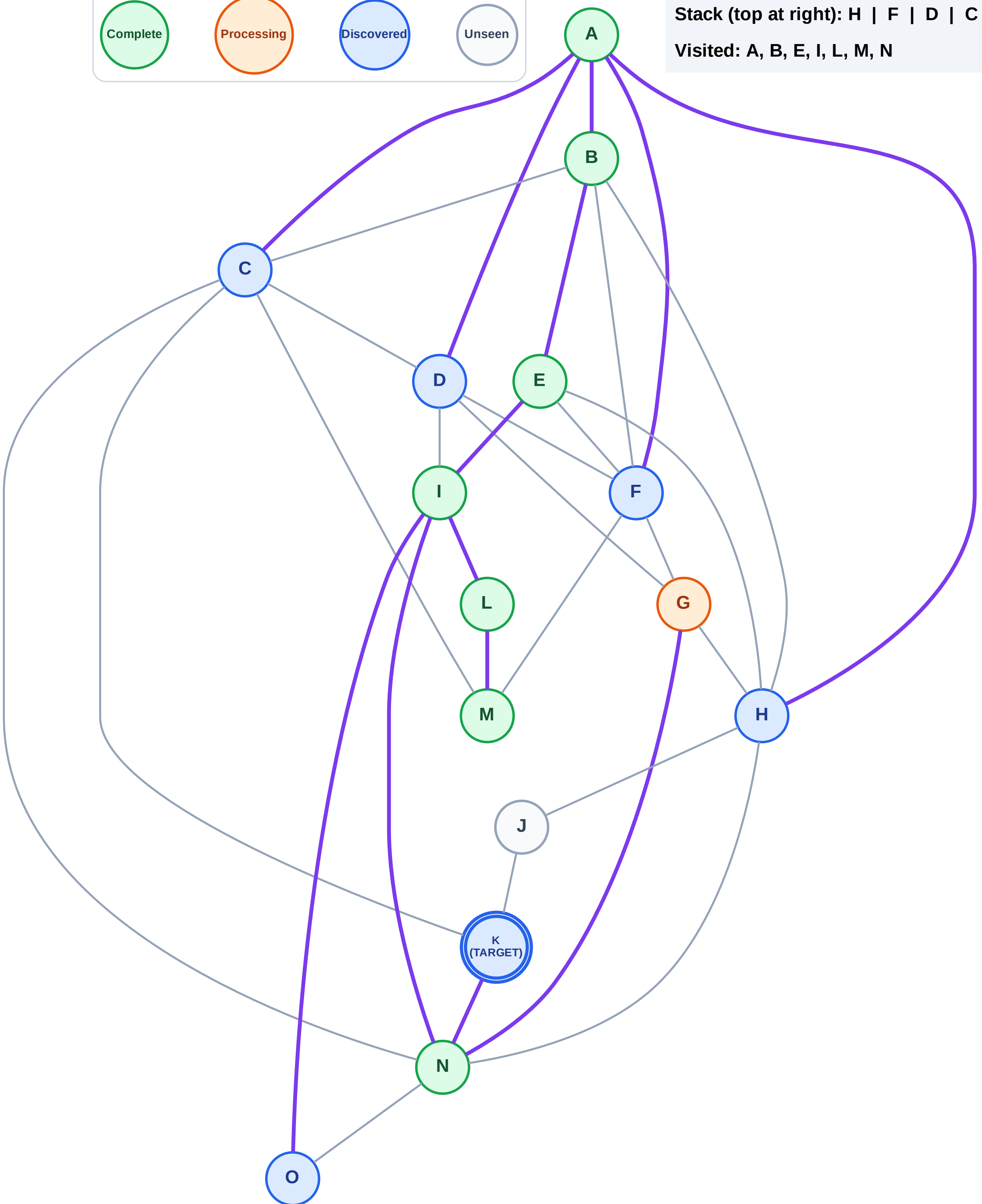
Step 16: Process node G

## Legend



Stack (top at right): H | F | D | C | O | K

Visited: A, B, E, I, L, M, N



# DFS Traversal

Step 17: No new neighbors from G

Legend

Complete

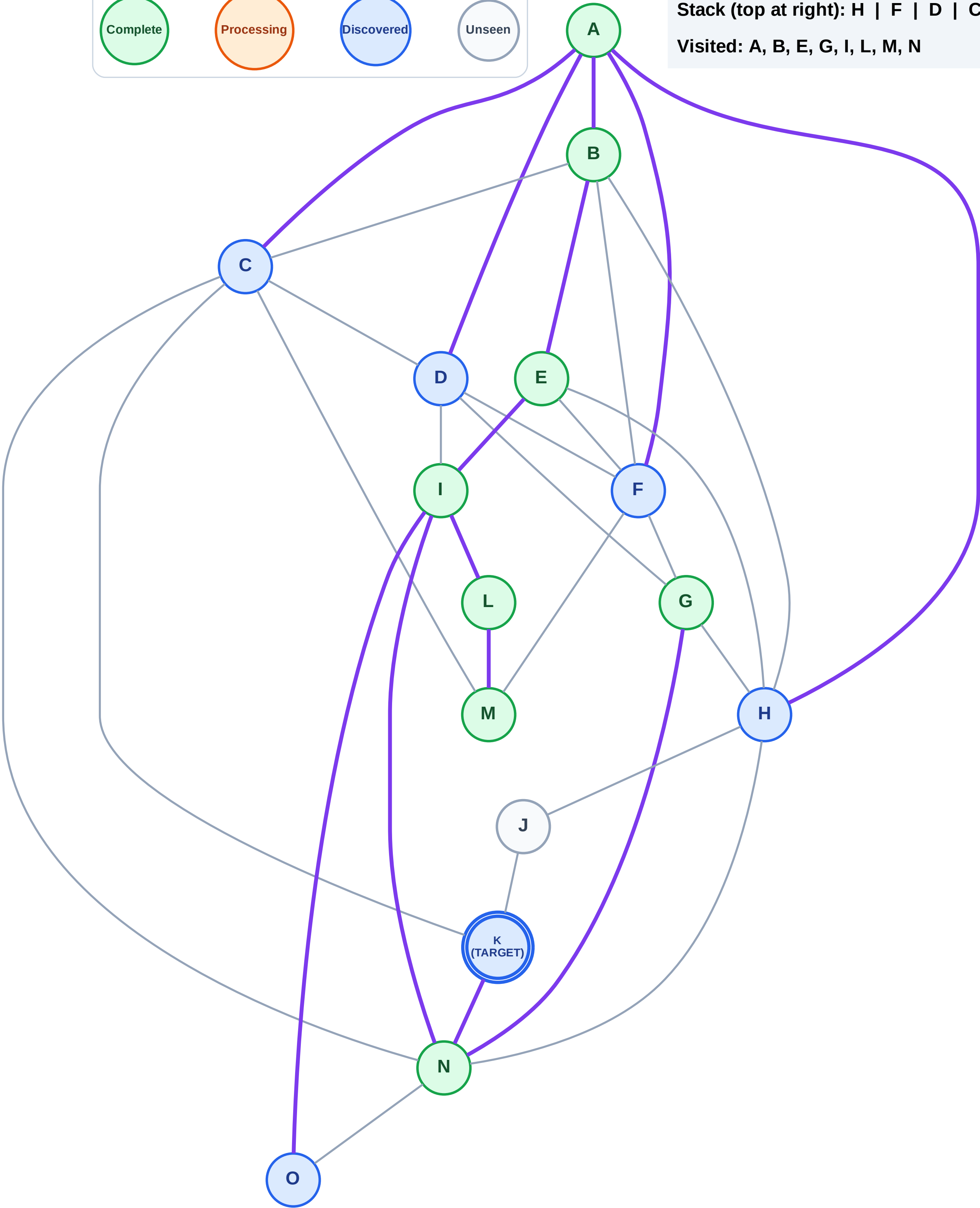
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | K

Visited: A, B, E, G, I, L, M, N



# DFS Traversal

Step 18: Process node K

Legend

Complete

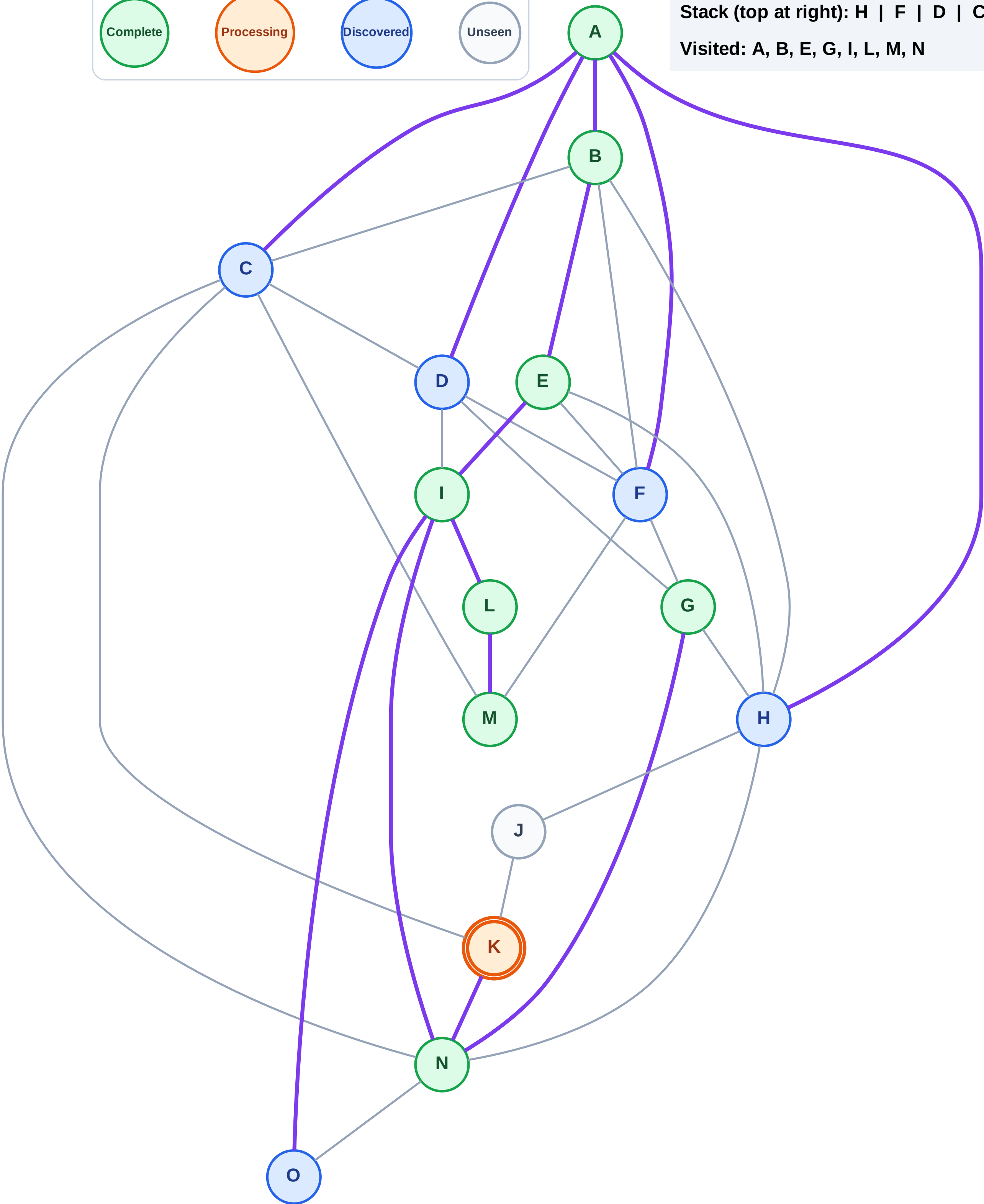
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O

Visited: A, B, E, G, I, L, M, N



# DFS Traversal

Step 19: TARGET NODE K REACHED! Path (5 hops): A → B → E → I → N → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O  
Visited: A, B, E, G, I, K, L, M, N

